Course Notes

Lecture 07 — ZigBee

Communication Technologies for the Internet of Things

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Text to support independent study

Contents

How to Use These Notes **2**

1 Commentary on the Slides **2**

2. Agenda (1/2) . . . 2
3. Agenda (2/2) . . . 3
4. ZigBee and the IEEE 802.15.4 Standard . . . 4
5. Boundary between IEEE 802.15.4 and ZigBee . . . 5
6. Terminology: IEEE 802.15.4 versus ZigBee . . . 6
7. General Characteristics of ZigBee (1/2) . . . 7
8. General Characteristics of ZigBee (2/2) . . . 8
9. Frequency Bands and Channels . . . 9
10. IEEE 802.15.4 Physical Layer . . . 10
11. Coexistence with Wi-Fi in the 2.4 GHz Band . . . 11
12. Physical Layer Packet Structure . . . 12
13. Device Types and Their Functions . . . 13
14. Device Roles: the Coordinator (1/2) . . . 14
15. Device Roles: the Coordinator (2/2) . . . 15
16. Device Roles: Router and End Device (1/2) . . . 16
17. Device Roles: Router and End Device (2/2) . . . 17
18. ZigBee Network: Devices and Links . . . 18
19. FFD, RFD and Modes of Operation (1/2) . . . 19
20. FFD, RFD and Modes of Operation (2/2) . . . 20
21. Medium Access: Superframe Structure . . . 21
22. Medium Access: CSMA/CA and Reliability . . . 22
23. The Three ZigBee Topologies . . . 23
24. Star Topology . . . 24
25. Tree Topology (Cluster-Tree) . . . 25
26. Mesh Topology . . . 26
27. Comparison of Topologies . . . 27
28. IEEE 802.15.4 Topologies . . . 28
29. Device Addressing (1/2) . . . 29
30. Device Addressing (2/2) . . . 30
31. Hierarchical Addressing and Tree Routing . . . 31

32. Mesh Routing: Route Discovery (AODV) . . . 32
33. Routing: Tables, Cost and Broadcast . . . 33
34. Network Formation and Associated Mechanisms (1/2) . . . 34
35. Network Formation and Associated Mechanisms (2/2) . . . 35
36. Joining a Device to the Network . . . 36
37. Indirect Delivery to Sleeping End Devices . . . 37
38. Protocol Stack and Responsibilities . . . 38
39. ZigBee Protocol Stack (1/2) . . . 39
40. ZigBee Protocol Stack (2/2) . . . 40
41. Frame Format and Encapsulation . . . 41
42. Security (1/2) . . . 42
43. Security (2/2) . . . 43
44. Application Profiles and ZigBee Device ID (1/2) . . . 44
45. Application Profiles and ZigBee Device ID (2/2) . . . 45
46. Data Model: Endpoint, Cluster and Attribute . . . 46
47. Energy Consumption (1/2) . . . 47
48. Energy Consumption (2/2) . . . 48
49. Duty Cycle and End Device Lifetime . . . 49
50. Typical Use Cases . . . 50
51. Integration in an IoT Architecture . . . 51
52. exlabel Home Automation: Lamps and Switches . . . 52
53. exlabel Smart Metering . . . 53
54. exlabel Smart Buildings and Lighting . . . 54
55. exlabel Agriculture, Industry and Healthcare . . . 55
56. exlabel ZigBee Application Families . . . 56
57. ZigBee Compared with Other Short-Range Technologies . . . 57
58. Advantages and Limitations (1/2) . . . 58
59. Advantages and Limitations (2/2) . . . 59
60. Evolution: ZigBee 3.0 and Matter . . . 60
61. Summary (1/2) . . . 61
62. Summary (2/2) . . . 62
63. Sources (1/2) . . . 63
64. Sources (2/2) . . . 64

Final Check **65**

References **65**

How to Use These Notes

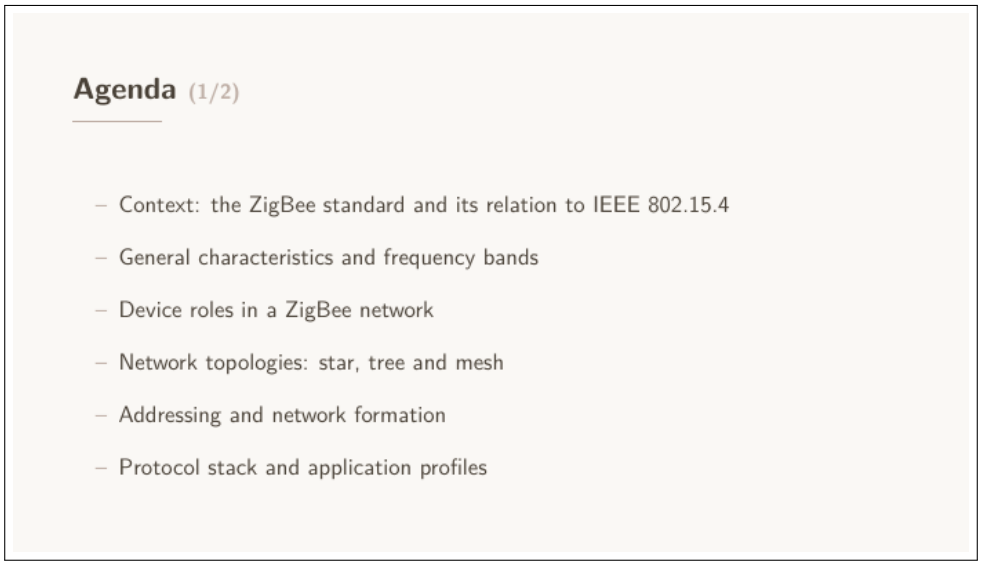
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

1. Commentary on the Slides

Slide 2. Agenda (1/2)



The lecture has one guiding question: what must be added to a low-power radio so that hundreds of devices form a reliable, secure and interoperable network. The agenda organises the answer in blocks that follow the order in which the technology is built, from the bottom up.

The first block places ZigBee relative to the IEEE 802.15.4 standard and fixes the general characteristics and frequency bands. The second presents the roles devices may have. The third, the topologies those roles make possible. The fourth deals with addressing and network formation, that is, how a new device becomes part of an existing network. The fifth goes through the protocol stack and the application profiles.

The agenda is best read as an index of dependencies. Routing cannot be understood without topologies, topologies without roles, or roles without the distinction between full-function and reduced-function devices. When studying, it is worth returning to the agenda and checking, for each block, which concepts depend on earlier blocks.

Slide 3. Agenda (2/2)

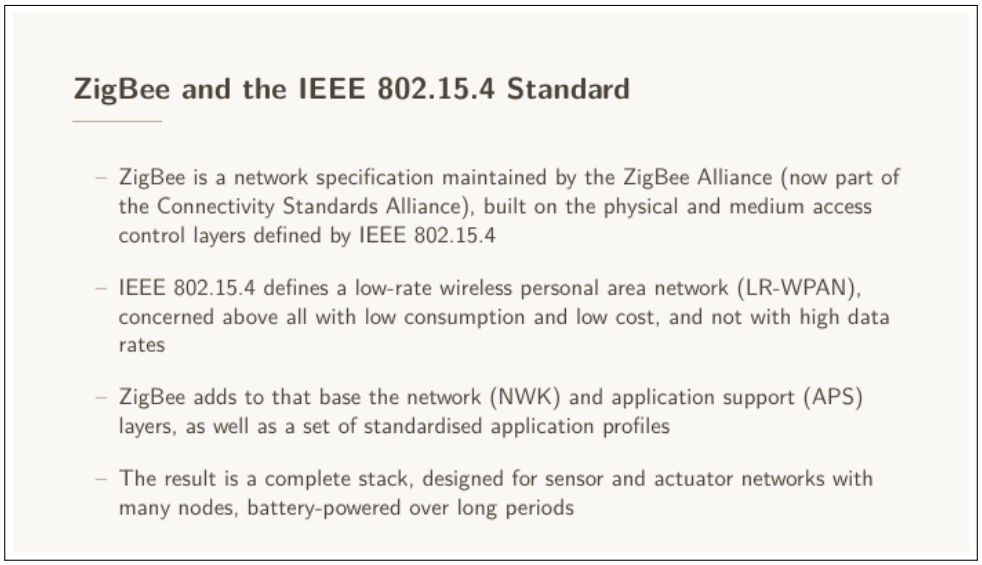


The second part of the agenda adds topics that are not strictly about networking but explain why the network has the form it has. The physical layer and medium access, routing, joining and security are the mechanisms that give substance to the topologies. Energy consumption and use cases show what all the rest is for.

Consumption is the criterion that determines almost every design decision in ZigBee: short frames, sleeping end devices, indirect message delivery and the requirement that routers be always on. The use cases, marked on the slides with a different background colour, tie those decisions to real installations.

The final summary recaps the concepts that should be fixed in memory. Throughout the slides, figures and tables carry the densest information; the bullet points explain them and the comments in these notes detail the reasoning.

Slide 4. ZigBee and the IEEE 802.15.4 Standard



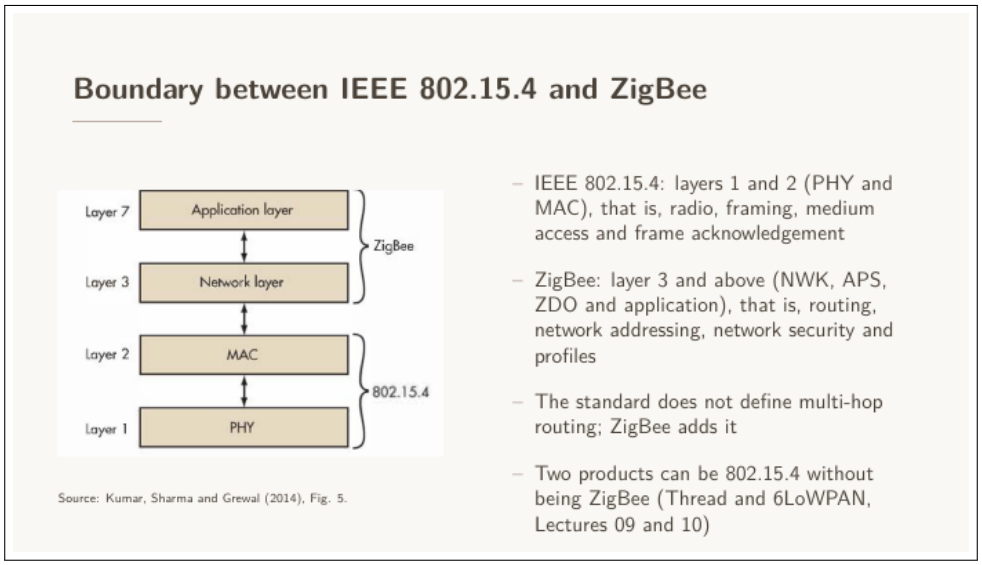
This is the framing slide, and the distinction it establishes runs through the whole lecture. IEEE 802.15.4 is a communications standard, maintained by the Institute of Electrical and Electronics Engineers, that describes a low-rate wireless personal area network (LR-WPAN) only at the two lowest levels of the reference model: the physical layer (PHY) and medium access control (MAC). ZigBee is a network and application specification, maintained by the ZigBee Alliance, now part of the Connectivity Standards Alliance (CSA), which uses 802.15.4 as a base and adds what is missing to form a complete network.

What is missing is substantial. 802.15.4 knows how to send a frame from one node to a neighbour and acknowledge reception, but it does not know how to route a message through several nodes, does not assign network addresses, does not define how a mesh topology is organised and does not standardise the meaning of the data exchanged. ZigBee adds the network layer (NWK), the application support layer (APS) and the application profiles.

The practical consequence is twofold. On the one hand, hardware and firmware can be shared among technologies that use the same radio, such as ZigBee and Thread. On the other, interoperability between manufacturers is only guaranteed above the MAC layer, by the standardised profiles.

Check question: which layers of the reference model are defined by the standard and which by the ZigBee specification?

Slide 5. Boundary between IEEE 802.15.4 and ZigBee

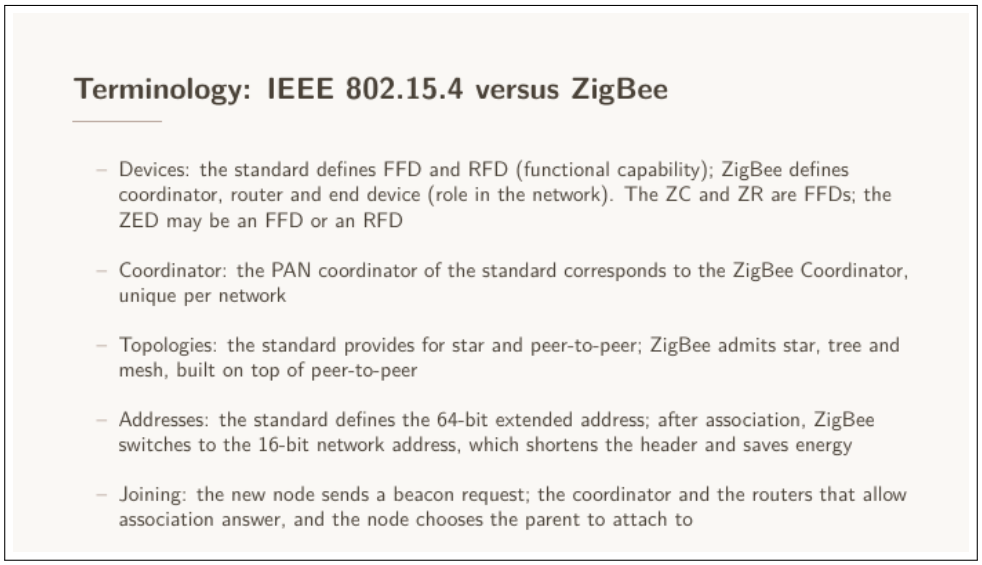


The figure, taken from the paper by Kumar, Sharma and Grewal (2014), summarises the boundary between the two standards. On the left are the numbered layers of the reference model; on the right, the braces show who owns each. Layers 1 and 2, PHY and MAC, belong to IEEE 802.15.4. Layer 3, the network layer, and the application layer belong to ZigBee.

The resulting table of responsibilities is as follows. The PHY handles radio activation, modulation, energy detection and clear channel assessment. The MAC handles channel access, framing, frame acknowledgement, association and beacon control. The NWK handles network addressing, routing and topology formation. The application layer, with the APS and the ZDO, handles service discovery, bindings and profiles.

The most commonly misunderstood point is that the 802.15.4 standard does not define multi-hop routing. With an isolated 802.15.4 radio, two nodes can only communicate if they are within range of each other. Routing through intermediaries is a creation of the ZigBee NWK layer. It also follows that other technologies use the same radio with a different stack, such as Thread and 6LoWPAN, studied in Lectures 09 and 10: they are 802.15.4 without being ZigBee.

Slide 6. Terminology: IEEE 802.15.4 versus ZigBee



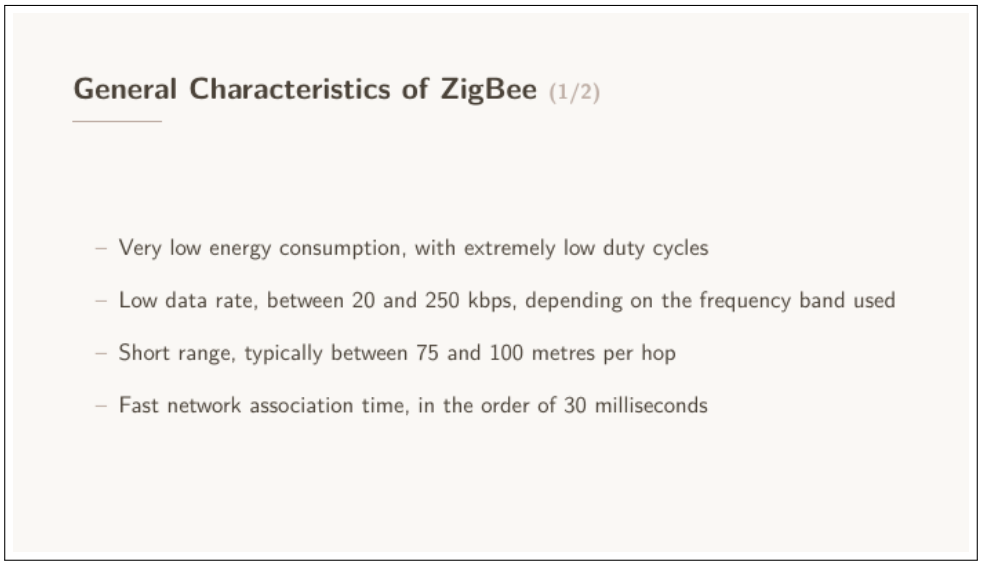
The slide takes inventory of the differences in vocabulary, which are a frequent source of confusion. The standard classifies devices by functional capability: full-function devices (FFD) implement the whole MAC and can play any role; reduced-function devices (RFD) implement a subset and can only talk to an FFD. ZigBee classifies devices by the role they play in the network: coordinator, router or end device. The correspondence is this: the coordinator and the router are always FFDs; the end device may be an FFD or an RFD.

The standard's PAN coordinator is the ZigBee coordinator, unique in each network. Topologies also differ: the standard provides only for the star and peer-to-peer, whereas ZigBee, built on peer-to-peer, admits star, tree and mesh.

The difference in addressing has energy consequences. Every device has a 64-bit extended address, globally unique. After association, ZigBee switches to the 16-bit network address assigned by the parent, which shortens each frame header by six bytes per address. Fewer bytes on the air means less transmission time and less energy.

Finally, joining the network. The new node sends a beacon request; the coordinator and the routers that allow association answer, and the new node chooses which parent to attach to. This logic is taken up again in the slide on network joining.

Slide 7. General Characteristics of ZigBee (1/2)



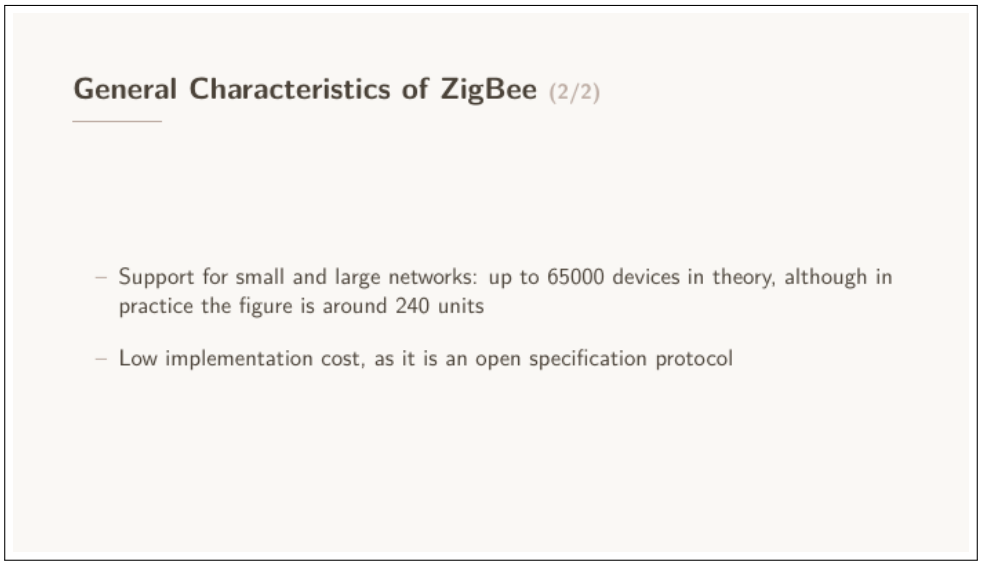
The slide gathers the numbers that characterise the technology, and they should be read as a set rather than in isolation. Consumption is very low, thanks to extremely low duty cycles: a device spends almost all its time at rest and switches on the radio only for milliseconds. The data rate is between 20 and 250 kbps, depending on the band, a modest value compared with Wi-Fi or even Bluetooth, and a deliberate one: to send a temperature reading or a switch-on command no more bits are needed, and fewer bits mean less energy.

The typical range is 75 to 100 metres per hop in favourable conditions. Indoors the value is lower, because walls and furniture attenuate the signal. This limitation is compensated by the mesh topology, in which each router adds a hop, so the area covered is well beyond the range of a single radio.

One characteristic that goes unnoticed is the association time, of the order of 30 milliseconds. A device that wakes, joins the network, sends its data and goes back to sleep spends much less energy than one that needs seconds to join.

The idea to retain is that ZigBee optimises energy, cost and scale, not data rate or range.

Slide 8. General Characteristics of ZigBee (2/2)

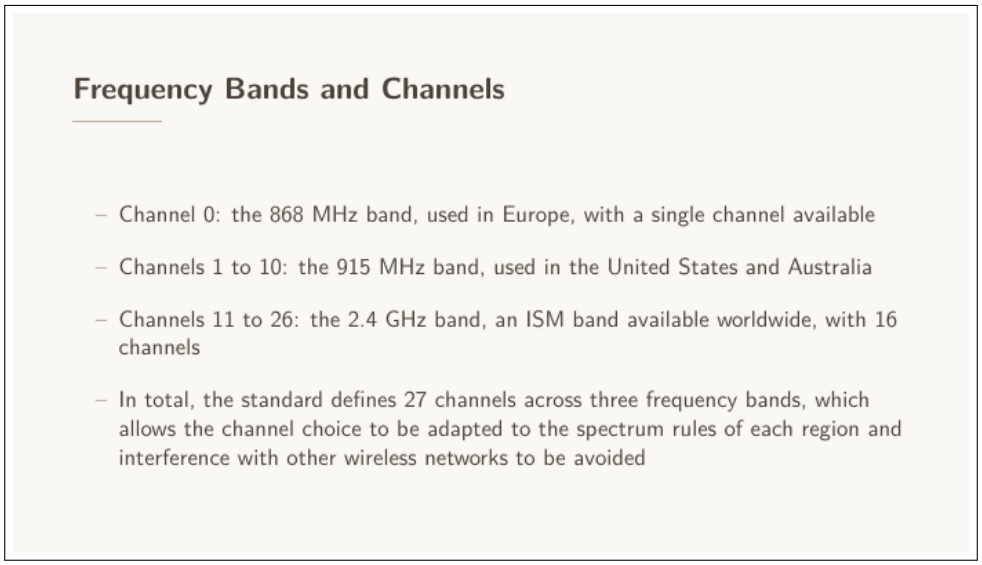


The standard allows, in theory, about 65 000 devices to be addressed per network, a number that follows from the 16-bit network address (65 536 values, some reserved). In practice, the number of nodes per network is around 240. The discrepancy is not a defect: it results from memory limitations in intermediate nodes, which must keep neighbour, routing and route discovery tables, and from the growth of control traffic with network size. A low-cost module has a few kilobytes of RAM and cannot hold thousands of entries.

In general, the effective size of a network is set by the scarcest resource: table memory, channel occupancy, or the user's tolerance of latency. That is why large installations are split into several networks, each with its own coordinator, connected through IP gateways.

Low implementation cost is the second characteristic on the slide. It follows from the specification being open, that is, any manufacturer can implement it without paying royalties for the protocol (certification and use of the trademark have their own costs), and from 802.15.4 hardware being produced in large volume. For IoT system design, the consequence is the possibility of installing many low-cost nodes, the opposite of the logic of a single expensive device.

Slide 9. Frequency Bands and Channels



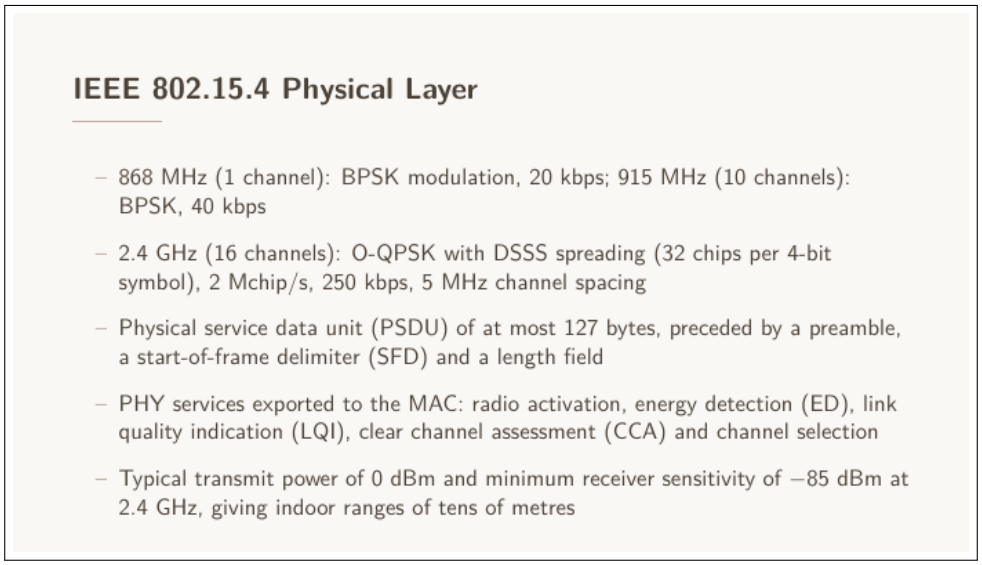
The standard defines 27 channels, distributed over three bands, and the choice of band depends on the region. Channel 0 is in the 868 MHz band, used in Europe, and is the only one in that band. Channels 1 to 10 are in the 915 MHz band, used in the United States and Australia. Channels 11 to 26 are in the 2.4 GHz band, an ISM band available worldwide, with 16 channels.

In practice almost all commercial products use 2.4 GHz, because a single product can be sold in all markets. The counterpart is that this is the most congested band: it is shared with Wi-Fi, Bluetooth, microwave ovens and many other sources. The sub-GHz bands are less crowded and propagate better, but the data rate is lower and the equipment must be adapted to the region.

The existence of many channels is what allows interference to be avoided. When forming the network, the coordinator measures the energy on each candidate channel and picks a free one. If interference appears later, the frequency agility mechanism, studied further on, allows the whole network to migrate to another channel.

Check question: why is the 2.4 GHz band predominantly used, despite being the most congested?

Slide 10. IEEE 802.15.4 Physical Layer



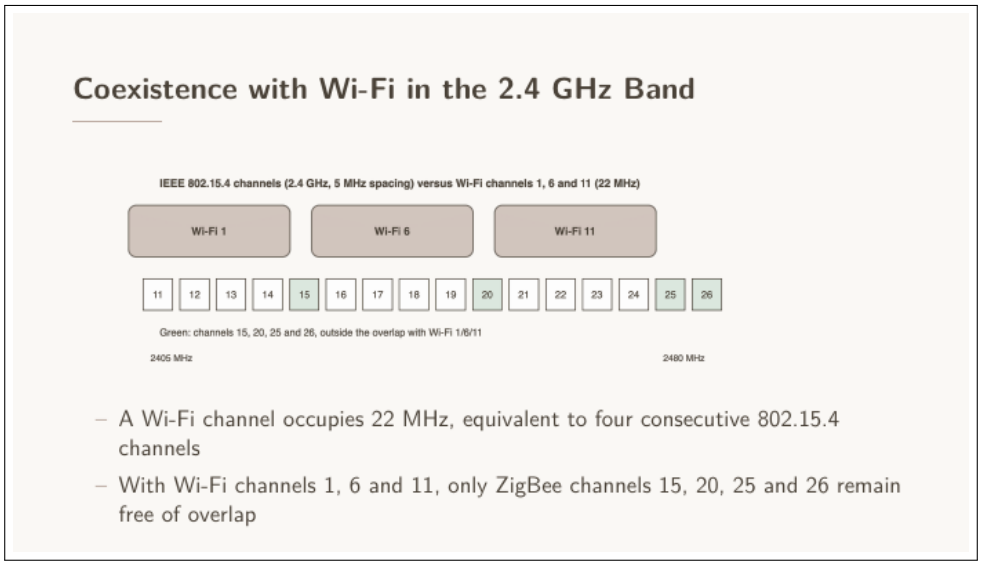
This slide goes down to the radio level. In each band the standard specifies a modulation and a data rate. In the 868 and 915 MHz bands BPSK is used, at 20 and 40 kbps respectively. In the 2.4 GHz band, O-QPSK with direct sequence spread spectrum (DSSS) is used: each 4-bit symbol is transmitted as a sequence of 32 chips, at 2 Mchip/s, giving 250 kbps of useful data rate. Spreading makes the link more robust to noise and interference at the cost of bandwidth: it is a deliberate trade of data rate for reliability.

Channels at 2.4 GHz are 5 MHz apart, and the spacing ensures neighbouring channels do not overlap. The data unit that the physical layer carries, the PSDU, has at most 127 bytes, which sets a hard limit on the size of any message and, consequently, on the application payload.

The physical layer provides the MAC with services that the upper protocols exploit. Energy detection (ED) measures the power on the channel and is used to choose a free channel. The link quality indication (LQI) characterises each received frame and is the raw material of the routing cost. Clear channel assessment (CCA) supports CSMA/CA.

With typical power of 0 dBm and minimum sensitivity of −85 dBm at 2.4 GHz, indoor ranges of tens of metres are obtained. Increasing the range requires more power, hence more energy, or more hops.

Slide 11. Coexistence with Wi-Fi in the 2.4 GHz Band

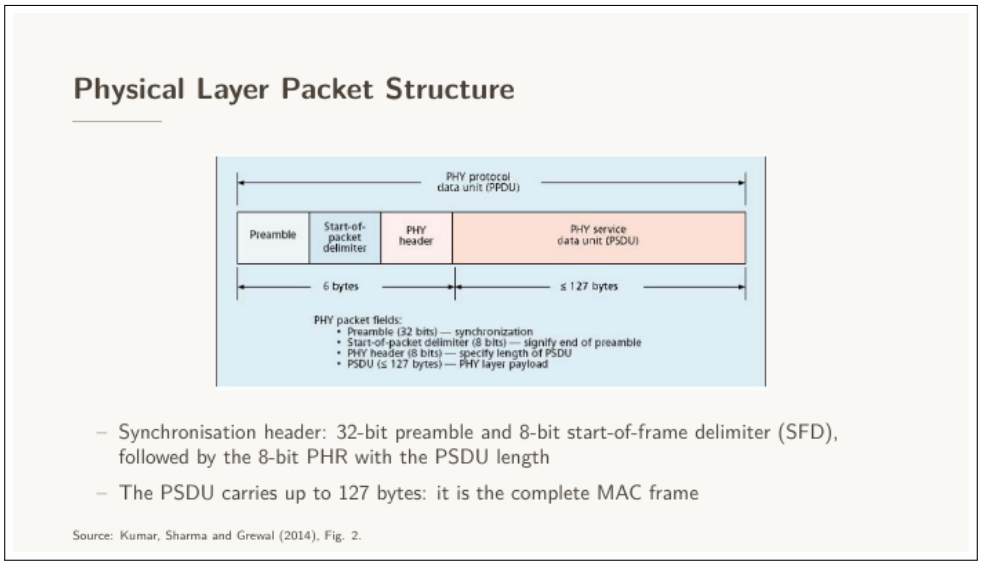


The chart shows the 16 channels of 802.15.4 in the 2.4 GHz band and the three most used Wi-Fi channels, 1, 6 and 11. A Wi-Fi channel occupies 22 MHz, so it covers the equivalent of four consecutive ZigBee channels. With Wi-Fi on channels 1, 6 and 11, which is the usual configuration precisely because they are the only three that do not overlap, ZigBee channels 11 to 14, 16 to 19 and 21 to 24 are overlapped. Channels 15, 20, 25 and 26 fall in the gaps and are preferred for ZigBee networks in environments with dense Wi-Fi.

There are three lessons from this slide. The first is that ZigBee and Wi-Fi compete for the same spectrum, and that Wi-Fi, with much higher transmit power, tends to prevail: ZigBee suffers more losses than it causes. The second is that channel choice is a design decision, not a detail: a badly chosen channel can degrade a whole network. The third is that coexistence is handled in planning, by assigning channels, and not only in the radio.

In Lecture 13, when configuring Zigbee2MQTT, the channel will be an explicit parameter, and the choice of channels 15, 20, 25 or 26 will be justified by this reasoning.

Slide 12. Physical Layer Packet Structure



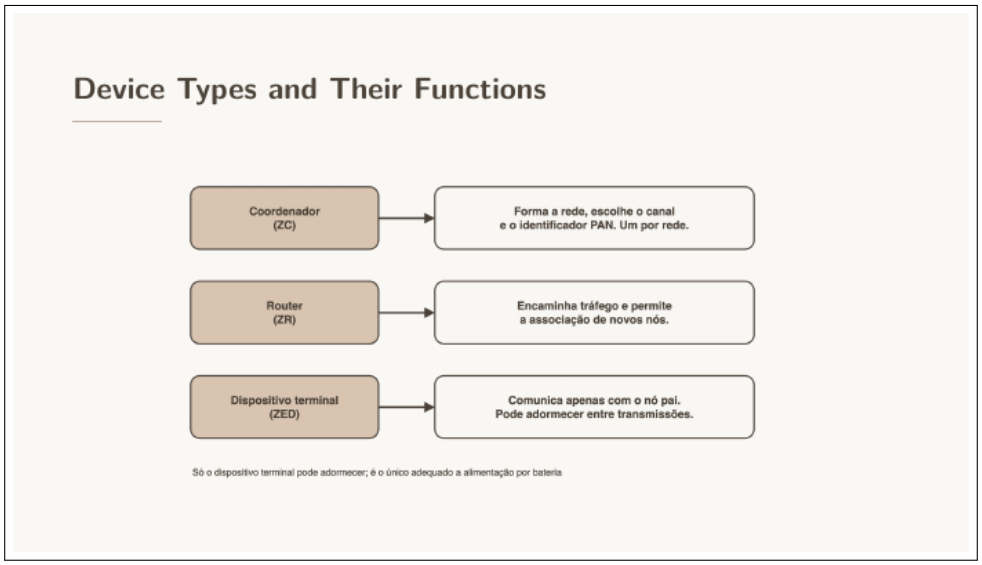
The figure, from Kumar, Sharma and Grewal (2014), shows the structure of the physical layer packet, which the standard calls the PPDU. It is what really travels through the air, and everything the upper layers build is inside its final part.

The packet starts with a synchronisation header, with the 32-bit preamble and the 8-bit start-of-frame delimiter (SFD). The preamble lets the receiver synchronise its clock and adjust its gain; the SFD marks the end of the preamble and the start of the frame proper. The 8-bit PHY header (PHR) follows, giving the length of the PSDU. These three fields add up to 6 bytes.

The PSDU, with a maximum of 127 bytes, is the complete MAC frame: MAC header, payload (which carries the ZigBee network layer frame) and the frame check sequence. The 127-byte limit is why ZigBee messages are short and why larger messages need fragmentation.

It is worth computing the cost. At 250 kbps, one byte takes 32 microseconds and a 127-byte frame plus the 6-byte PHY header occupies the channel for about 4.3 milliseconds. Airtime is directly radio-on time, hence energy consumed: that is why every header byte counts.

Slide 13. Device Types and Their Functions



This is the first of the slides that present the three roles in a ZigBee network, and the figure is the visual summary to return to. The three device types are the coordinator (ZC), the router (ZR) and the end device (ZED).

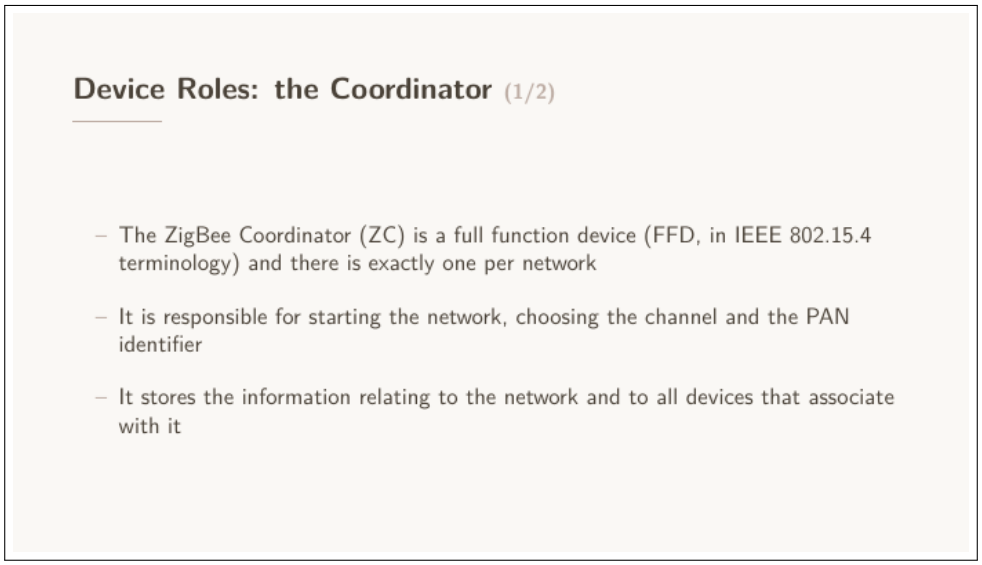
The coordinator forms the network and there is exactly one per network. The router forwards messages between nodes and allows other devices to associate to it; it is optional. The end device is the leaf of the network: it communicates with its parent, does not forward third-party traffic and may sleep.

The figure associates with each role its properties: functional capability (FFD or RFD), type of power supply (mains or battery), ability to sleep, and function. The relation between role and power is the key. A device that forwards must always be listening, because it does not know when traffic from another node will arrive; hence it cannot sleep and in practice must be mains-powered. A device that only talks to its parent can agree with it on the moments when it wakes, and so can run on battery for years.

This asymmetry is the ZigBee design principle: concentrate the energy cost of connectivity in the nodes that have permanent electricity and free the rest.

Check question: why can only the end device be battery powered?

Slide 14. Device Roles: the Coordinator (1/2)



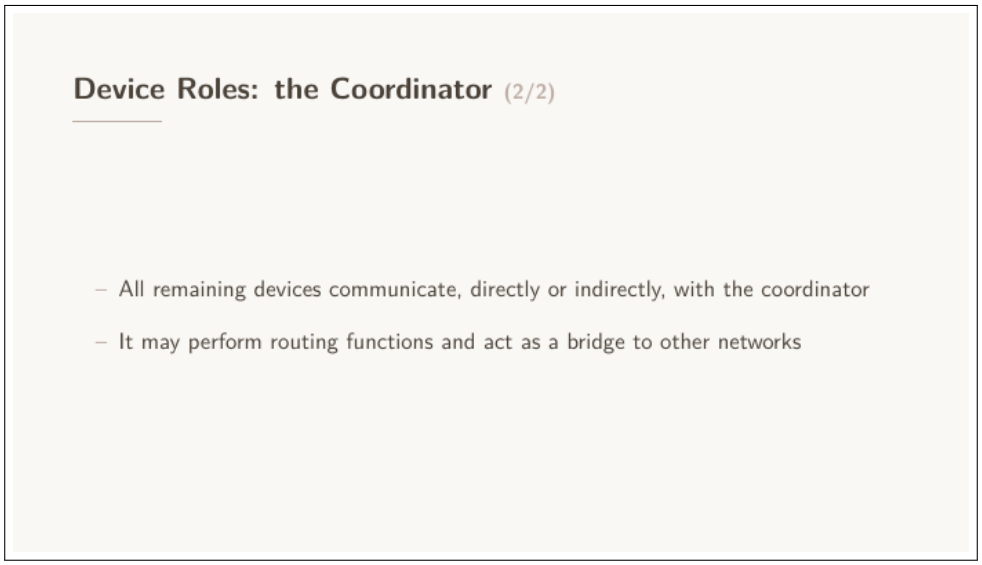
The ZigBee coordinator (ZC) is a full-function device and there is exactly one per network. Its first responsibility is to start the network. On start-up it scans the candidate channels for energy, picks one with little interference and also chooses the PAN identifier (PAN ID), 16 bits long, which distinguishes its network from neighbouring networks sharing the same space. From then on the network exists and accepts associations.

The second responsibility is to keep the network information: the list of associated devices, the assigned addresses, the network key and, in many implementations, the role of Trust Center, which decides who may enter and distributes the security keys.

It is important to stress what the coordinator is not. It is not a central server through which all traffic passes: in mesh networks, traffic between two routers does not pass through it. Its distinctive function is exercised in network formation and security, not in day-to-day routing. This explains an operational fact: in many mesh networks, once formed, the coordinator can be switched off temporarily without interrupting communication among the other nodes, although no new devices can join.

When the network is a star, the situation changes: the coordinator is then the single point of passage and the single point of failure.

Slide 15. Device Roles: the Coordinator (2/2)

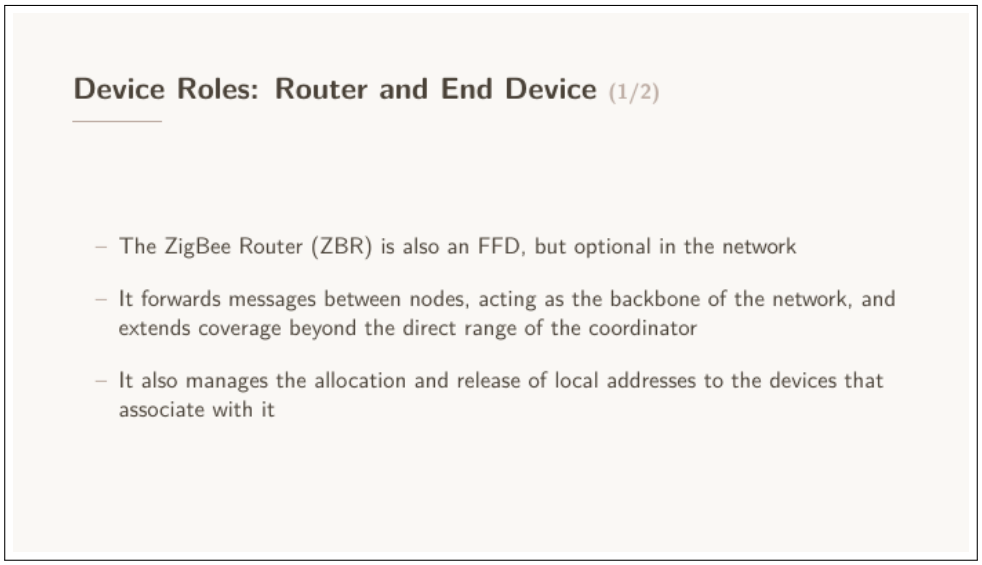


The second half of the coordinator description explains its place in the system. All other devices communicate with it, directly or indirectly: directly, when within range of its radio; indirectly, when communication goes through intermediate routers. This formulation matters because it describes the network as a tree of associations rooted at the coordinator, even when actual routing is a mesh.

The coordinator may also perform routing functions and act as a bridge to other networks. This last role anticipates the gateway studied in the BLE lectures and which will be central in Lecture 13: in practice, the ZigBee dongle plugged into a Raspberry Pi is the coordinator, and the program running on the Raspberry Pi translates what it receives into MQTT messages. From the IoT architecture point of view, the coordinator is therefore the point where the ZigBee sensor network meets the IP network.

Design concerns follow. Being unique and holding the network state, the coordinator is an element to protect: it is advisable to power it permanently, back up its configuration, and, in consumer products, it is known that replacing the hub often implies re-pairing every device.

Slide 16. Device Roles: Router and End Device (1/2)



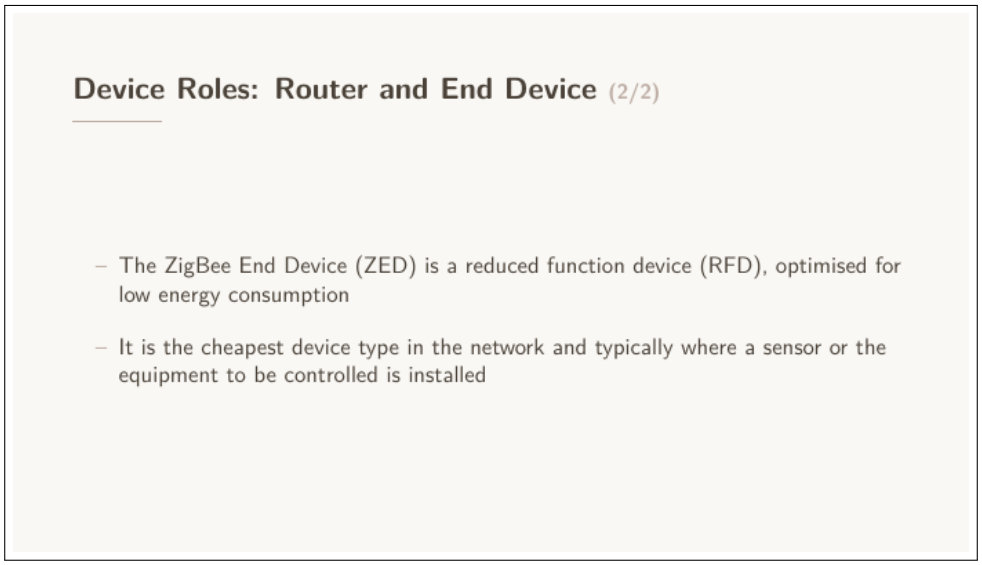
The ZigBee router (ZR) is also a full-function device, but its presence is optional. A minimal network can consist of a coordinator and end devices connected directly to it, in a star. Routers add capacity and range.

The router's main function is to forward messages between nodes. It receives frames from a neighbour and retransmits them towards the destination, working as the backbone of the network. In doing so it extends the coverage area beyond the coordinator's direct radio range: each router adds a hop and, with it, a new circle of coverage. A long corridor or a multi-storey building, impossible to cover with a single short-range radio, becomes coverable with a few well-placed routers.

The second function is to manage the assignment and release of local addresses to the devices that associate with it. When a new node chooses a router as its parent, that router gives it the 16-bit network address and keeps the information about its presence. The router is thus an entry point to the network, just like the coordinator.

Because of these functions the router must always be active. In real installations, lamps, smart plugs and meters play the router role, because they have permanent power and are distributed through the space.

Slide 17. Device Roles: Router and End Device (2/2)



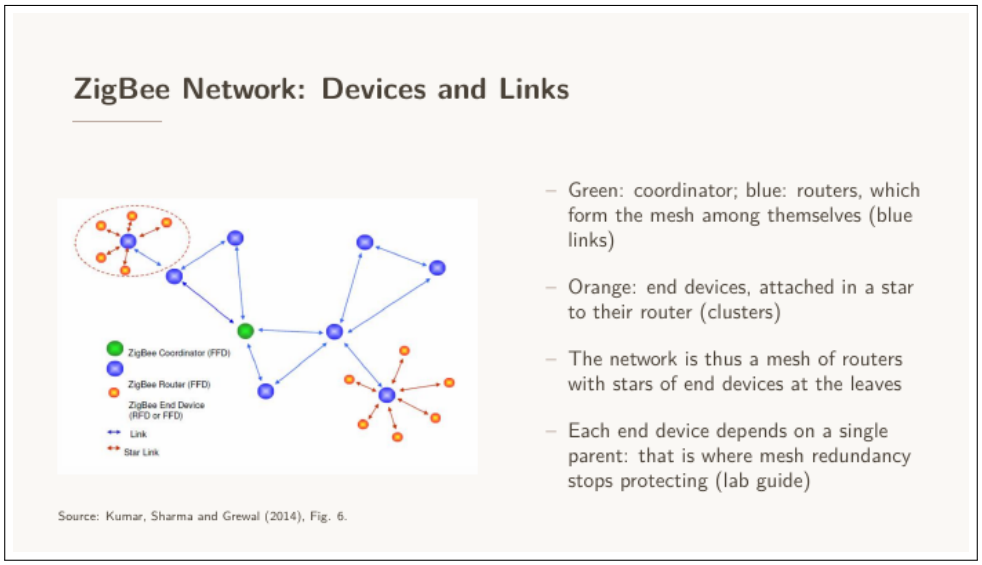
The ZigBee end device (ZED) is reduced-function, or, more precisely, may be: nothing prevents an FFD from behaving as an end device. What defines it is behaviour, not hardware: it communicates only with its parent, does not forward third-party messages and may sleep.

This simplicity brings two advantages. The first is cost: without routing tables or router functions, the module needs less memory and is cheaper, making it the most economical device type in the network. The second is energy autonomy: since it does not have to listen permanently, the end device spends most of its time in a rest mode, with consumption of the order of a microampere, and switches on the radio only to send a reading or check whether the parent has messages waiting. With a coin cell, autonomy can reach years.

The end device is where the sensor or the equipment to be controlled is typically installed: the temperature sensor, the door contact, the button, the motion sensor. In an installation with many nodes, most devices are end devices.

There is, however, a structural consequence that will be explored in the lab guide: the end device depends on a single parent. If that parent fails, the end device is isolated, however redundant the router mesh above it may be.

Slide 18. ZigBee Network: Devices and Links



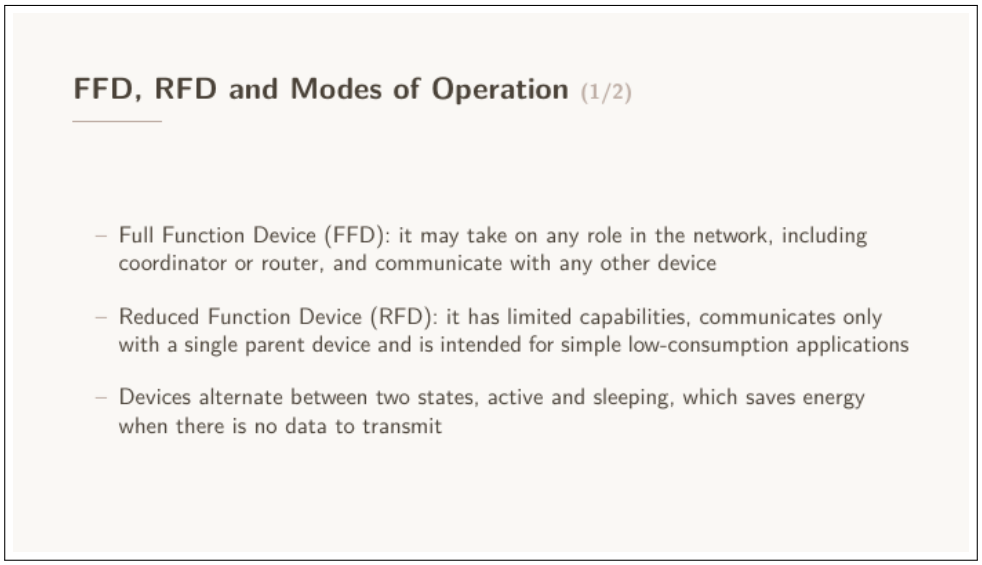
The figure, from Kumar, Sharma and Grewal (2014), shows a real ZigBee network in abstract form and is a good synthesis of the three roles in one drawing. The green circle is the coordinator. The blue circles are the routers, linked to each other and to the coordinator by blue arrows representing mesh links. The orange circles are end devices, attached in a star to the router they belong to; the dashed circle highlights one such grouping.

The overall structure is a mesh of routers with stars of end devices at the leaves. This is the topology real ZigBee networks end up with, even when they are described as star, tree or mesh. The central part, of routers, is meshed and redundant; the periphery, of end devices, is a star without redundancy.

This observation has consequences for reliability. Mesh redundancy protects communication among routers: if one fails, traffic goes around it. But it does not protect the end devices that depended on that router as their parent, because each end device has only one parent. In the lab activity, the failure of ZR1 isolates ZED1 and ZED2 even in the mesh network, and this figure shows why.

In design, this weak point is mitigated by placing enough routers that an end device has several candidate parents, and by letting an end device that has lost its parent re-associate with another.

Slide 19. FFD, RFD and Modes of Operation (1/2)



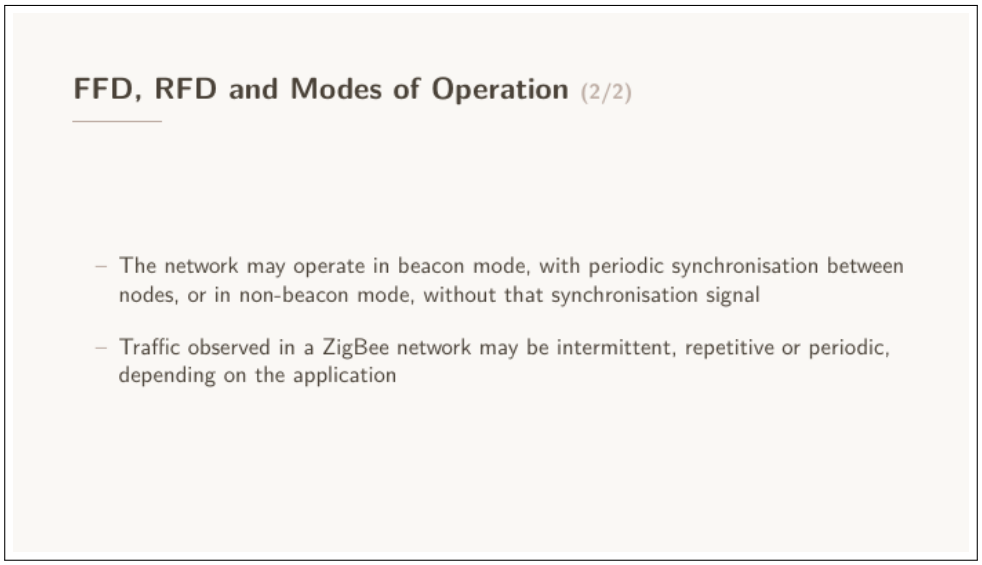
The slide presents two device classes, defined by the 802.15.4 standard, and should be read together with the earlier slide on terminology. A full-function device (FFD) can play any role, including coordinator or router, and communicates with any other device. A reduced-function device (RFD) has limited capabilities, communicates only with a single parent device and is meant for simple low-power applications, such as a switch or a sensor.

The distinction is one of capability, not of function. An FFD can behave as an end device, if configured so. The reverse is impossible: an RFD cannot be a router, because it does not implement the necessary functions.

The second point on the slide is that devices alternate between two states, active and sleeping. When active, a device draws thousands of times more current than when asleep. Energy savings therefore come from maximising rest time. This is the duty-cycle principle, which will be quantified on a later slide.

The sleeping state has a cost: while asleep, the device does not receive. Any message addressed to it has to wait. That problem is solved by indirect delivery, in which the parent holds the messages until the child wakes and asks for them.

Slide 20. FFD, RFD and Modes of Operation (2/2)



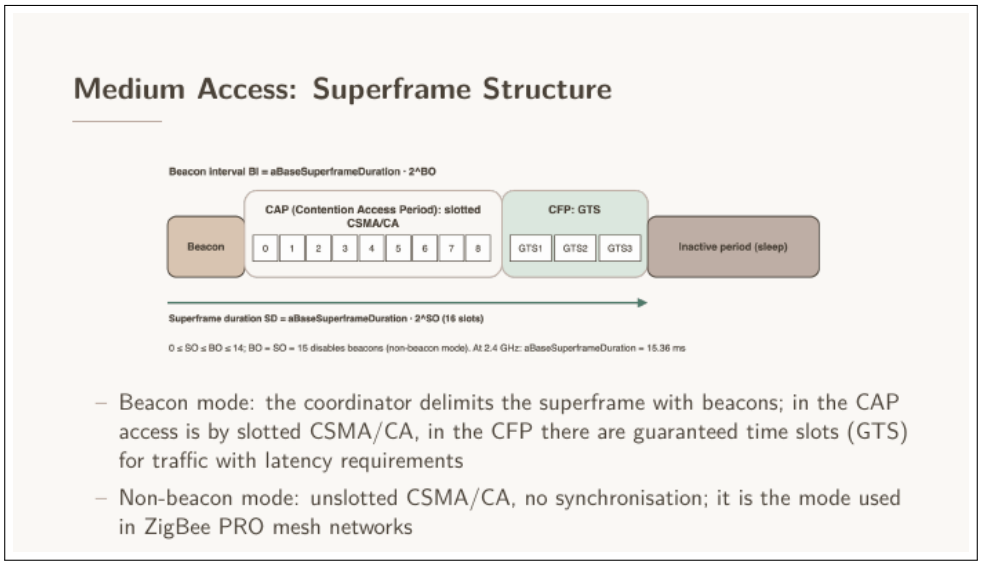
The second slide on operating modes introduces the distinction between beacon mode and non-beacon mode, which stems from the 802.15.4 MAC. In beacon mode, the coordinator (or router) periodically emits a special frame, the beacon, which serves as a time reference for the whole network. Devices synchronise to it and know when they may transmit and when they may sleep. In non-beacon mode there is no such synchronisation: each device accesses the channel on its own, after checking that it is free.

The choice is a trade-off. Beacon mode allows guaranteed times and very low duty cycles, but requires nodes to keep their clocks synchronised and limits network scale. Non-beacon mode is more flexible and is the one used in mesh networks, because routers, which must always listen, do not need synchronisation to forward.

The slide also mentions the traffic pattern, which can be intermittent (event-triggered, such as an alarm), repetitive (such as a control command) or periodic (such as a sensor reading at fixed intervals). Each pattern suggests a configuration. Periodic traffic favours beacon mode and scheduled sleeping; intermittent traffic favours non-beacon mode and immediate response.

The following slides study the time structure of beacon mode, the superframe.

Slide 21. Medium Access: Superframe Structure



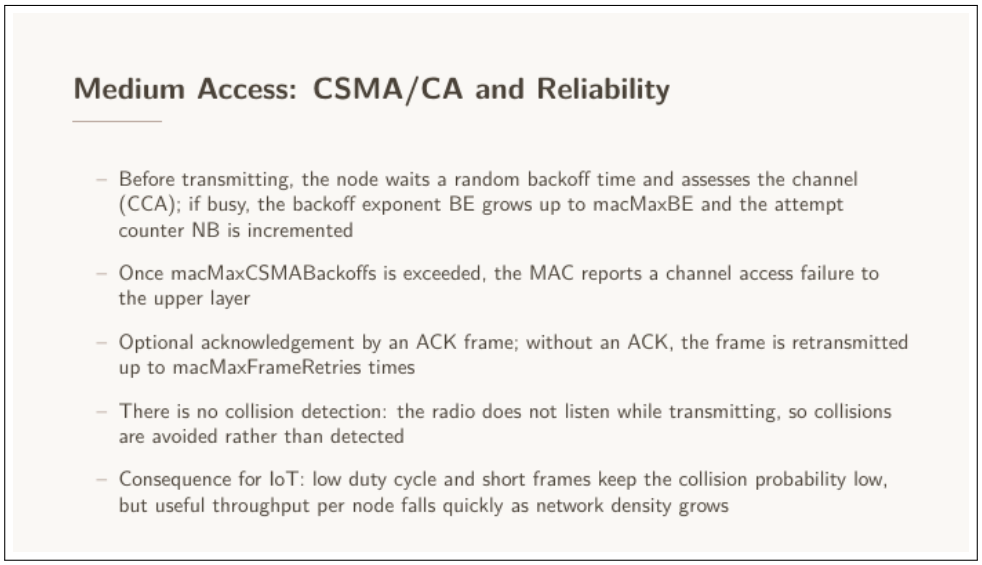
The figure represents the IEEE 802.15.4 superframe in beacon mode. Time is divided into beacon intervals, each started by a beacon emitted by the coordinator. Within each interval, the superframe has two parts: an active period, divided into 16 slots, and, optionally, an inactive period in which all devices may sleep.

The active period splits into two phases. In the contention access period (CAP), devices compete for the channel by slotted CSMA/CA: each waits for the start of a slot before trying. In the contention-free period (CFP) there are up to seven guaranteed time slots (GTS), which the coordinator assigns to devices with latency requirements so they can transmit without competition.

Two parameters define the structure. The beacon order (BO) sets the interval between beacons, BI = aBaseSuperframeDuration $\cdot 2^{BO}$. The superframe order (SO) sets the duration of the active period, SD = aBaseSuperframeDuration $\cdot 2^{SO}$, with $0 \leq SO \leq BO \leq 14$. The base duration is 960 symbols, or 15.36 milliseconds at 2.4 GHz. If $SO < BO$, the remainder of the interval is inactive and the network sleeps. The network duty cycle is thus 2^{SO-BO}.

Example: with $BO = 6$ and $SO = 2$, the interval is about 983 milliseconds and the active period about 61 milliseconds, a duty cycle of 6.25 per cent. The values $BO = SO = 15$ disable beacons, which corresponds to the non-beacon mode used by ZigBee PRO in mesh networks.

Slide 22. Medium Access: CSMA/CA and Reliability



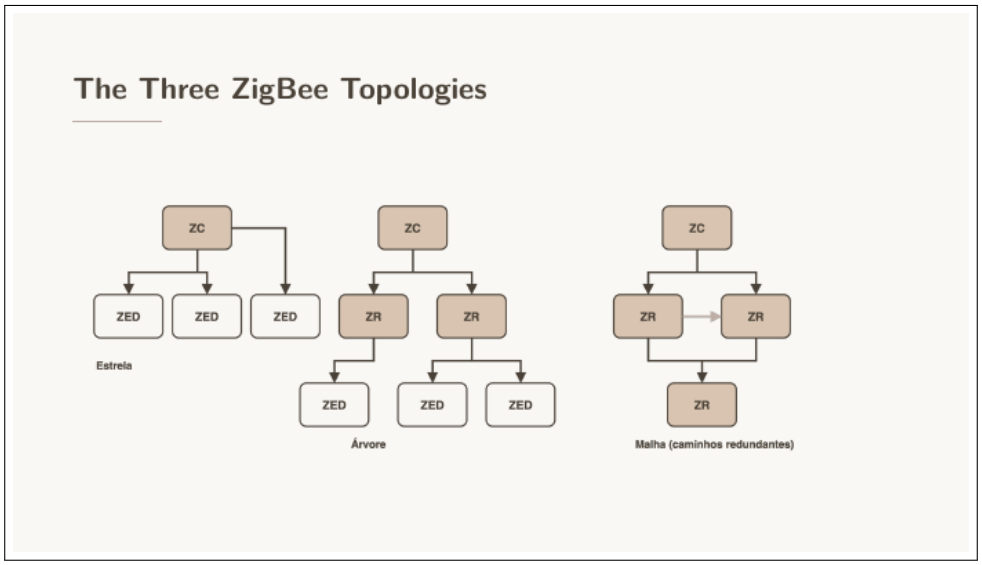
The slide describes the mechanism by which devices compete for the shared medium, CSMA/CA, short for carrier sense multiple access with collision avoidance. The decisive word is avoidance: unlike classic Ethernet, which detects collisions while transmitting, the 802.15.4 radio cannot listen while it transmits, and so it tries to avoid the collision rather than detect it.

The algorithm is simple. Before transmitting, the node waits a random time, the backoff, chosen from an interval whose size is set by the exponent BE. After the backoff, it assesses the channel (CCA). If free, it transmits. If busy, it increases the exponent BE, widening the choice interval, increments the attempt counter NB and repeats. When NB exceeds macMaxCSMABackoffs, the MAC gives up and informs the upper layer of the channel access failure.

Reliability is treated separately. If the sender asks for acknowledgement, the receiver answers with an ACK frame; in the absence of an ACK, the sender retransmits up to macMaxFrameRetries times. The two mechanisms should be distinguished: CSMA/CA prevents collisions up front, while acknowledgement with retransmission recovers losses that occur anyway, through interference or fading.

The consequence for IoT is that performance depends on density. With few nodes and short frames, the collision probability is low. As density grows, backoffs, retransmissions and latency increase, and the useful throughput per node drops quickly. It is one of the reasons for the practical limit on nodes per network.

Slide 23. The Three ZigBee Topologies



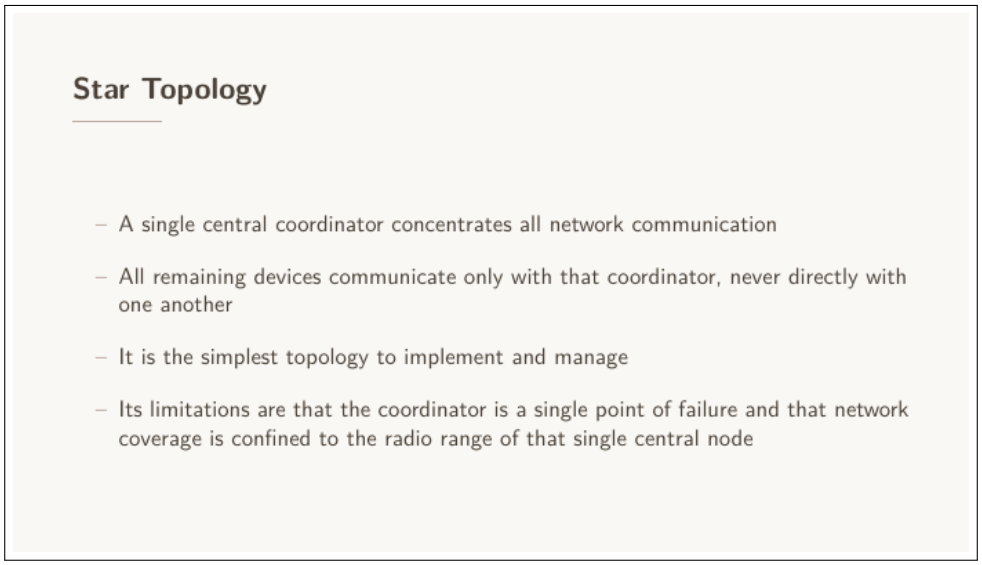
The figure compares the three ZigBee topologies side by side and is the reference slide for what follows. In the star, all devices connect directly to the coordinator. In the tree, routers create hierarchical branches from the coordinator. In the mesh, there are additional links between routers, shown dashed, which create alternative paths.

It should be noted that the three topologies are not different networks but configurations of the same technology. The star is the particular case without routers. The tree adds routers with a rigid parent-child linking rule. The mesh lifts that rule and lets routers communicate with any neighbour in range.

What the figure shows, and the text of the following slides explains, is also of interest. The star has a single point of failure and range limited to that of the coordinator. The tree extends range, but each branch depends on its router. The mesh adds redundancy: if a router fails, traffic goes around it.

The comparative study of the topologies is the core of the lab activity, in which a model of the network is used to measure which nodes become isolated when a router fails. The quantitative result, and not just the qualitative statement, is what supports the conclusion about robustness.

Slide 24. Star Topology



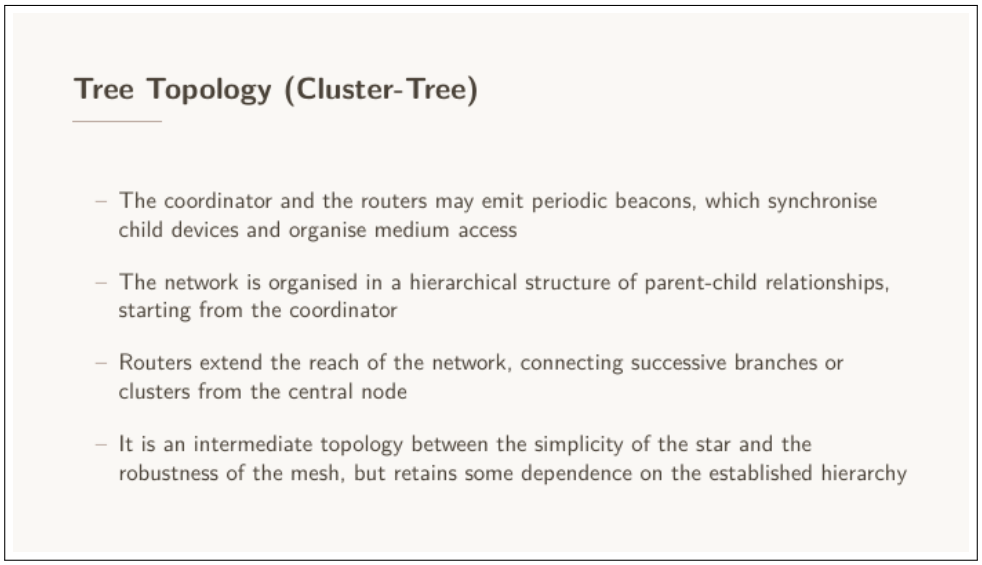
In the star topology there is a single central coordinator, which concentrates all communication in the network. All other devices communicate only with that coordinator, never directly with each other. If a sensor wants to send a message to a lamp, the path is sensor, coordinator, lamp: two hops, even if the two devices are side by side.

The advantage is simplicity. There are no routing tables to maintain or routes to discover, latency to the coordinator is minimal (one hop) and behaviour is predictable. For a small installation, with all devices within range of the coordinator, it is the natural choice, and it is also the easiest topology to debug.

The limitations are two and are on the slide. The first is that the coordinator is a single point of failure: if it fails, the whole network stops working. The second is that coverage is confined to the radio range of that single central node; a more distant device cannot join the network. Adding devices also increases the load on the coordinator and contention on the channel.

From the point of view of the 802.15.4 standard, the star is the native topology, with the PAN coordinator at the centre and reduced-function devices at the ends. From the ZigBee point of view, it is the most elementary case, which is outgrown when more area must be covered or failures tolerated.

Slide 25. Tree Topology (Cluster-Tree)



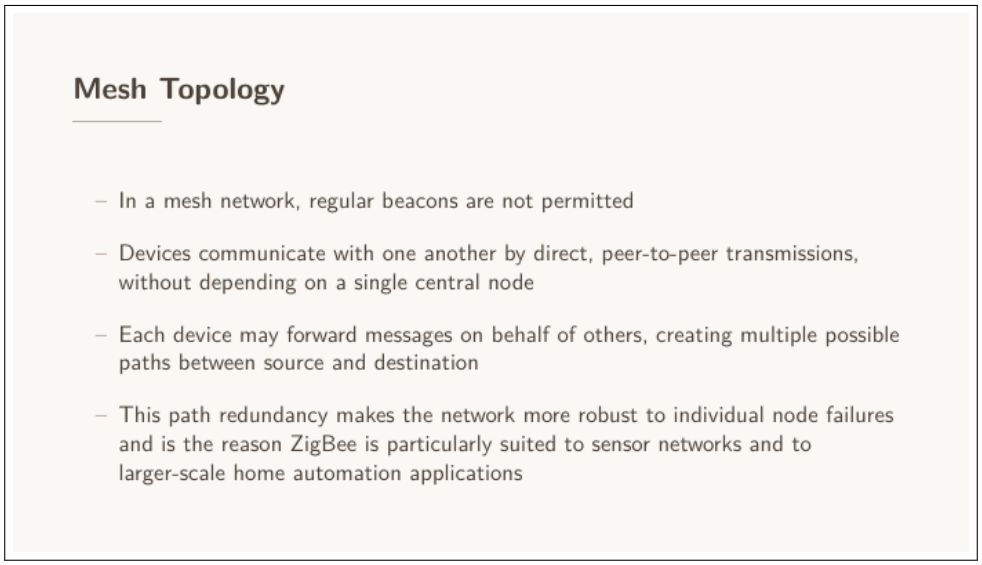
The tree topology, or cluster-tree, organises the network in a hierarchical structure of parent-child relationships, starting at the coordinator. The coordinator has children; each child that is a router may in turn have its own children, and so on up to a maximum depth. Routers extend the reach of the network, linking successive branches, or clusters, from the central node.

In the original variant, the coordinator and routers may emit periodic beacons, which synchronise their children and organise medium access. Each router defines its own superframe, scheduled so as not to collide with its parent's.

Tree routing is deterministic and needs no tables: each node knows the address block assigned to each branch and decides whether to forward the packet to a child, if the destination is in its block, or to the parent, otherwise. The advantage is predictability and memory saving; the drawback is that the path between two neighbouring nodes can be long, because it has to climb to the common root and descend.

The weakness is the dependence on the hierarchy. The failure of a router isolates the whole subtree that depended on it, and there is no alternative path. That is why the tree is described as an intermediate topology: it has more range and scale than the star, but less robustness than the mesh. In the lab guide, the tree network is obtained by removing the redundant links from the mesh, and the effect on the failure of ZR1 and ZR2 is immediate.

Slide 26. Mesh Topology



In the mesh topology, devices communicate with each other by direct transmissions, peer to peer, without depending on a single central node. The slide says regular beacons are not allowed, which means the mesh uses non-beacon mode: there is no global synchronisation, each node accesses the channel by CSMA/CA and routers are permanently listening.

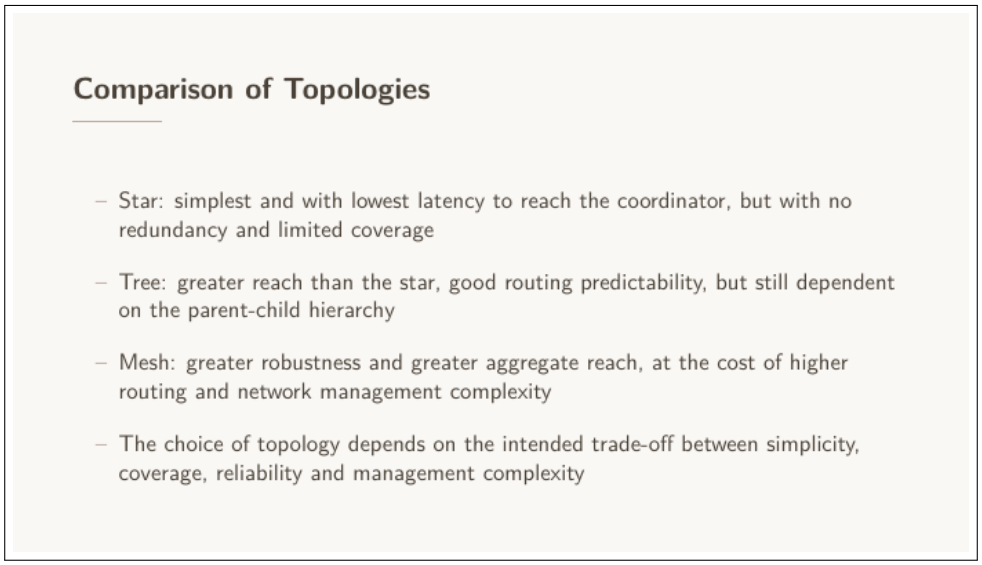
The essential property is path redundancy. Each router can forward messages on behalf of others, and there is typically more than one path between source and destination. If a router fails, or the link between two routers degrades, the network discovers another path and carries on without user intervention. This automatic recovery capability (self-healing) is what makes ZigBee suitable for sensor networks and larger-scale home automation.

The price of the mesh is complexity. A route discovery protocol, routing tables, discovery tables, link quality management and controlled broadcast mechanisms are needed. These mechanisms consume memory and generate control traffic, and are studied in the slides on routing.

It is worth stating what the mesh does not solve. Redundancy protects the paths between routers, not the end devices that depended on a router that failed, because each end device has only one parent.

Check question: what happens to the network when a router fails, in tree and in mesh topology?

Slide 27. Comparison of Topologies



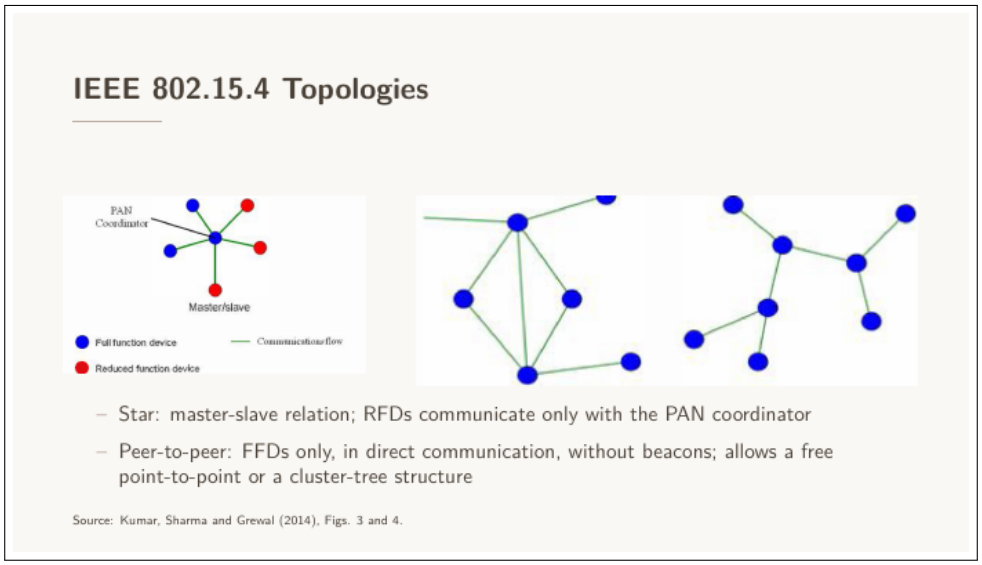
The slide summarises the comparison of the topologies on four criteria: simplicity, latency, robustness and range. The star is the simplest and has the lowest latency to the coordinator, but has no redundancy and limited coverage. The tree has greater range than the star and good routing predictability, but remains dependent on the parent-child hierarchy. The mesh has the greatest robustness and the greatest aggregate range, at the cost of greater routing and management complexity.

No topology is globally better. The choice depends on the desired trade-off among simplicity, coverage, reliability and management complexity. An installation of ten devices in one room may be very well served by a star. An installation of thirty devices on three floors needs routers to cover the area and, if availability matters, mesh redundancy.

This is also the basis of the discussion question in the lab activity, which asks for a justified recommendation for a home automation installation. The answer is not found by memorisation but by reasoning about the requirements: range, fault tolerance, latency, number of nodes, energy budget.

In short, the star trades robustness for simplicity, the tree trades robustness for predictability, and the mesh trades simplicity for robustness.

Slide 28. IEEE 802.15.4 Topologies



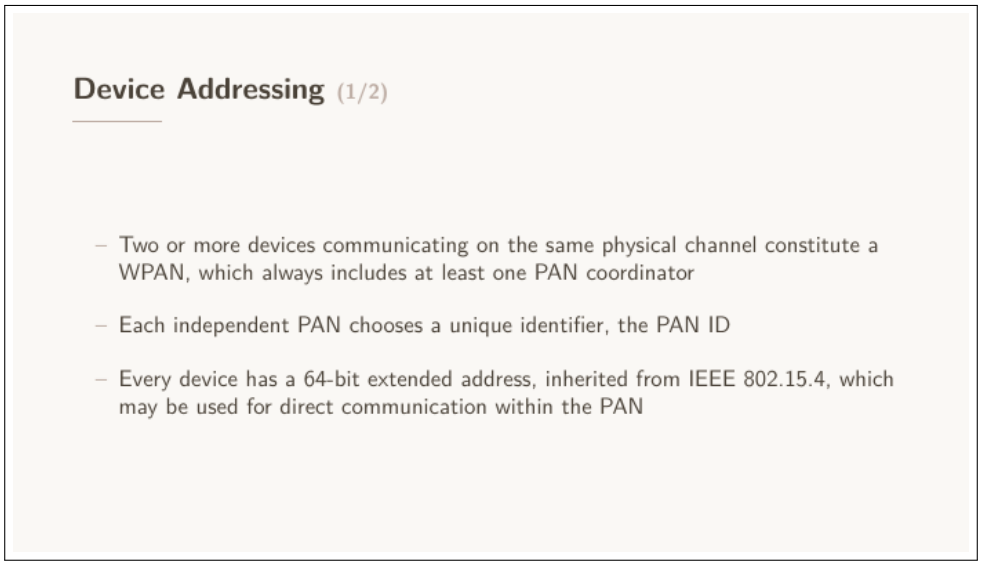
The two figures, from Kumar, Sharma and Grewal (2014), represent the topologies that the IEEE 802.15.4 standard itself defines: star and peer to peer. They should be distinguished from the ZigBee topologies, which are built on top of them.

In the standard's star, the full-function device at the centre is the PAN coordinator, in a master-slave relation with the others. Blue dots represent FFDs, red dots RFDs, and green lines the communication flow. RFDs communicate only with the coordinator and never with each other.

In peer to peer, only FFDs take part, communicating directly with one another, with no master-slave relation and no beacons. The figure shows two forms: free point to point, on the left, in which each node links to whoever is in range, and cluster-tree, on the right, in which the network is organised in hierarchical branches.

The correct reading is this: the standard's peer to peer offers the physical possibility of multi-hop communication, but does not say how to route. ZigBee uses that possibility and adds, in the network layer, the addressing and routing that make the mesh a usable network. That is why ZigBee is said to be implementable on any topology, whereas the standard alone describes only star and peer to peer.

Slide 29. Device Addressing (1/2)



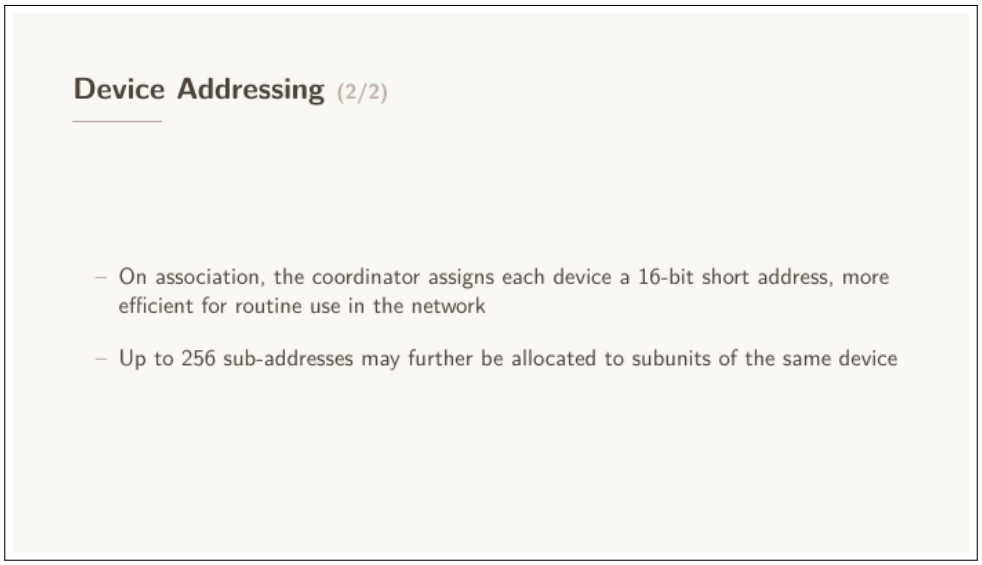
Two or more devices communicating on the same physical channel form a wireless personal area network (WPAN), which always includes at least one PAN coordinator. Each independent PAN chooses a unique identifier, the PAN ID, which allows neighbouring networks, on the same or adjacent channels, to coexist without confusing each other's frames: a node ignores frames whose PAN ID differs from its own.

Every device has a 64-bit extended address (EUI-64), inherited from IEEE 802.15.4, assigned at the factory and globally unique. It is the equivalent of the MAC address of a network card. It may be used for direct communication within the PAN, but it is long: eight bytes per address, which in a frame with source and destination addresses take sixteen of the 127 bytes available.

The solution is the second address, presented on the next slide, of 16 bits. The reason two addresses exist is the difference between identity and location. The 64-bit address identifies the device, is permanent and serves, for example, to recognise it when it re-joins. The 16-bit address identifies the device's position in that network, is assigned at association and may change.

Check question: why does each device have two addresses, one of 64 bits and one of 16 bits?

Slide 30. Device Addressing (2/2)



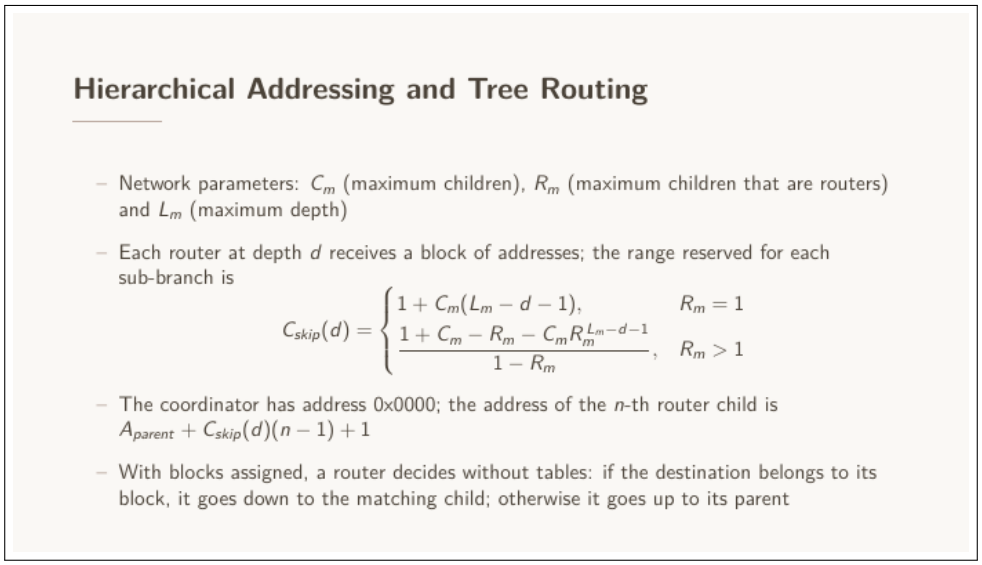
At association, the coordinator, or the router acting as parent, assigns the device a 16-bit short address. This address is used in the day-to-day life of the network, because it shortens each frame header and, with it, reduces transmission time and consumption. The coordinator always has address 0x0000.

Up to 256 sub-addresses, the endpoints, may also be allocated to sub-units of the same device. A physical module with a lamp and a temperature sensor is a single network device, with one address, but two endpoints. Endpoint 0 is reserved for the ZigBee device object (ZDO), endpoints 1 to 240 are for applications, and 255 is used for broadcast to all endpoints.

In sum, a ZigBee message is addressed by three values: the PAN ID, which chooses the network, the network address, which chooses the device, and the endpoint, which chooses the application within the device. This hierarchy recalls that of an IP network, with the IP address and the transport port, but is applied in a much lighter stack.

ZigBee also supports group addressing and broadcast. A group allows a single command to be sent to several devices, for example, switching on all the lamps in a room, saving traffic.

Slide 31. Hierarchical Addressing and Tree Routing



The slide presents the distributed hierarchical addressing of the original ZigBee specification, and it is the only one in which mathematics is unavoidable. The idea is to assign each router a contiguous block of addresses, so that routing can be done without tables, only by comparing addresses.

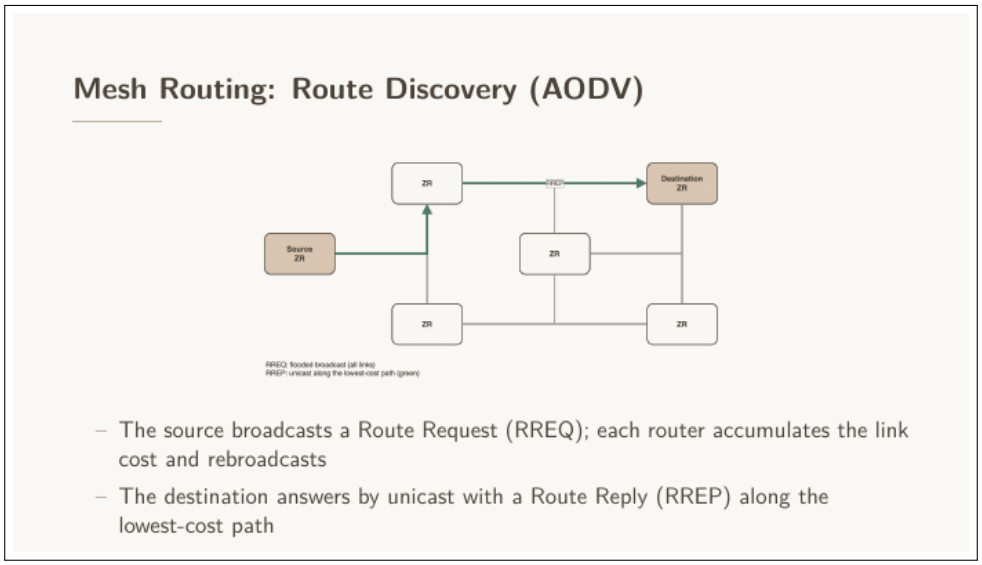
There are three parameters. C_m is the maximum number of children of a node, R_m the maximum number of those children that may be routers, and L_m the maximum depth of the tree. A router at depth d reserves, for each router child, a block of size $C_{skip}(d)$, given by the expression on the slide. If $R_m = 1$ the progression is linear; otherwise it is geometric.

The coordinator has address 0x0000 and the n-th router child of a node with address A at depth d receives $A + C_{skip}(d)(n-1) + 1$. Example: with $C_m = 4$, $R_m = 4$ and $L_m = 3$, $C_{skip}(0) = (1 + 4 - 4 - 4 \cdot 4^2)/(1 - 4) = 21$. The first router child of the coordinator has address 1, the second 22, the third 43.

How routing is done: a router with address A and depth d receives a packet for destination D. If $A < D < A + C_{skip}(d-1)$, the destination is in its subtree and the packet goes down to the child whose block contains it; otherwise it goes up to the parent. No tables are needed.

The limitation is evident: the parameters are fixed and must be known beforehand. Reserved but unused blocks waste address space. This rigidity led to the introduction of stochastic addressing in ZigBee PRO.

Slide 32. Mesh Routing: Route Discovery (AODV)



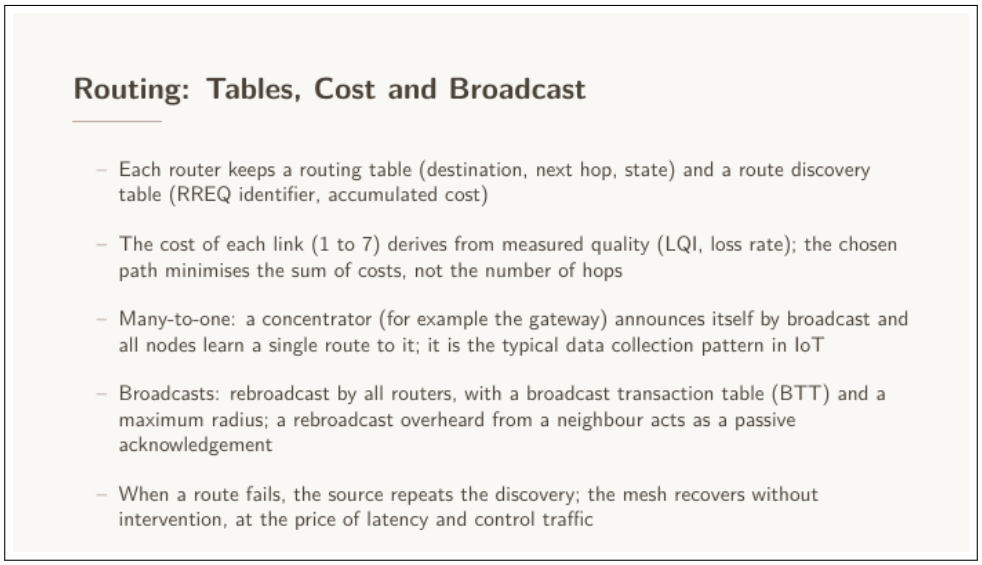
The figure illustrates route discovery in a mesh, which ZigBee inherits from the AODV algorithm (Ad hoc On-demand Distance Vector). The key phrase is "on demand": routes are only discovered when needed and are not kept permanently, which saves memory and traffic.

When a source router wants to send to a destination whose path it does not know, it broadcasts a route request (RREQ) to all neighbours. Each router that receives it adds to the accumulated cost the cost of the link it arrived on and rebroadcasts it. Because the request is flooded, it reaches the destination over several paths, and each router records, in the route discovery table, the best cost seen for that request, discarding the worse ones.

The destination answers with a route reply (RREP) sent by unicast, along the lowest-cost path, in the reverse direction. Along that path, each router records in the routing table the next hop towards the destination. In the end, the source knows where to send the packet and the intermediate routers know how to forward it. In the figure, the green path is the result.

Three remarks. Broadcast has a cost: each RREQ occupies the channel across the whole network, so routes should be kept and reused. The chosen path minimises cost and not the number of hops, because cost reflects link quality. And a route may expire or fail, in which case the discovery is repeated.

Slide 33. Routing: Tables, Cost and Broadcast



The slide gathers the operating details of mesh routing. Each router keeps two tables. The routing table records, for each known destination, the next hop and the route state. The route discovery table temporarily stores the requests in progress, with the RREQ identifier and accumulated cost, and allows duplicate and worse RREQs to be discarded.

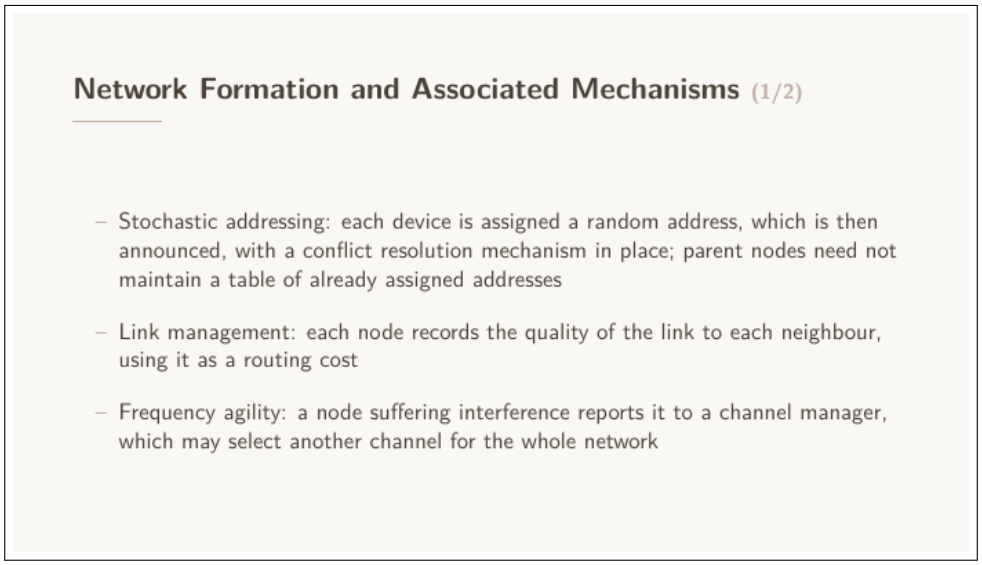
The cost of each link, from 1 to 7, derives from the measured quality, from the LQI and the loss rate. A good-quality link has cost 1; a weak link, a higher cost. The chosen path minimises the sum of costs. A three-hop path with good links may therefore be preferred to a two-hop path with a marginal link, which is correct, because the marginal link would generate retransmissions, delay and consumption.

The many-to-one pattern is typical of data collection. A concentrator, for example the gateway, announces itself by broadcast and all nodes learn a single route to it. Instead of each sensor discovering its route to the gateway, one broadcast serves the whole network.

Broadcasts are retransmitted by all routers, with a broadcast transaction table (BTT) to avoid loops and a maximum radius to limit reach. A rebroadcast overheard from a neighbour acts as a passive acknowledgement, dispensing with explicit ACKs.

When a route fails, the source repeats the discovery. The mesh recovers without intervention, at the cost of latency and control traffic. It is the counterpart of robustness.

Slide 34. Network Formation and Associated Mechanisms (1/2)



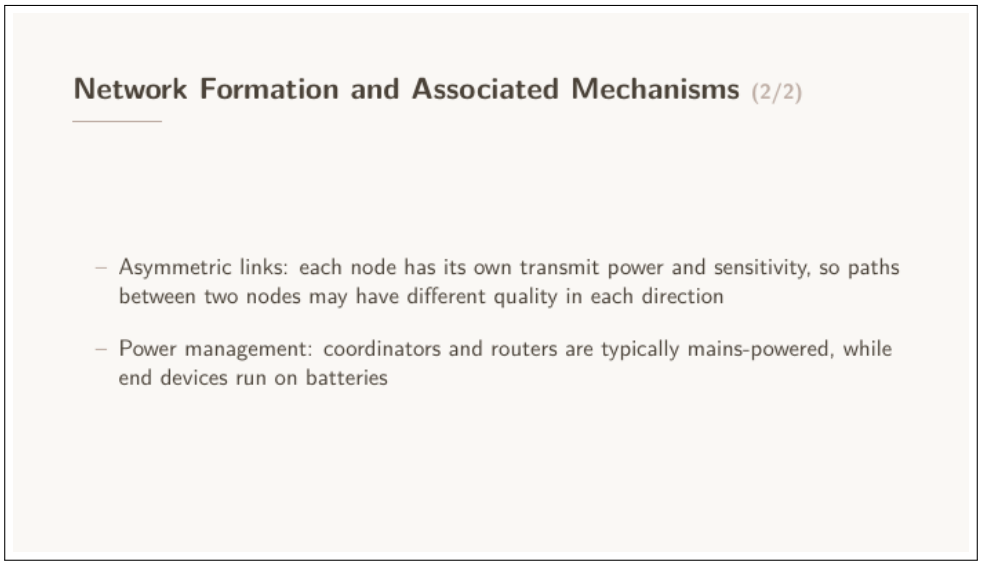
The slide presents the first mechanisms of network formation and maintenance. Stochastic addressing, introduced in ZigBee PRO, resolves the rigidity of hierarchical addressing: each device is assigned a random 16-bit address, which is then announced to the network. If two devices choose the same address, there is a mechanism to detect and resolve the conflict, in which one of them chooses another. The great advantage is that parent nodes need not keep a table of addresses already assigned nor reserve blocks, and the network can grow without prior planning. The drawback is that tree routing, which relied on blocks, is no longer possible, and mesh routing with tables becomes mandatory.

Link management records the quality of the link to each neighbour, measured from received frames, and uses it as the routing cost. Each router periodically exchanges link status messages with its neighbours and keeps a neighbour table with the cost of each. This is the information AODV uses to choose paths.

Frequency agility is the third mechanism. A node suffering persistent interference reports it to a channel manager, usually the coordinator, which may decide to move the whole network to another channel. All devices are notified and migrate together. It solves prolonged interference, such as a neighbouring Wi-Fi network that starts operating, but does not solve node failures: if a router breaks, changing channel does not repair it.

Check question: what is frequency agility and what problem does it solve?

Slide 35. Network Formation and Associated Mechanisms (2/2)

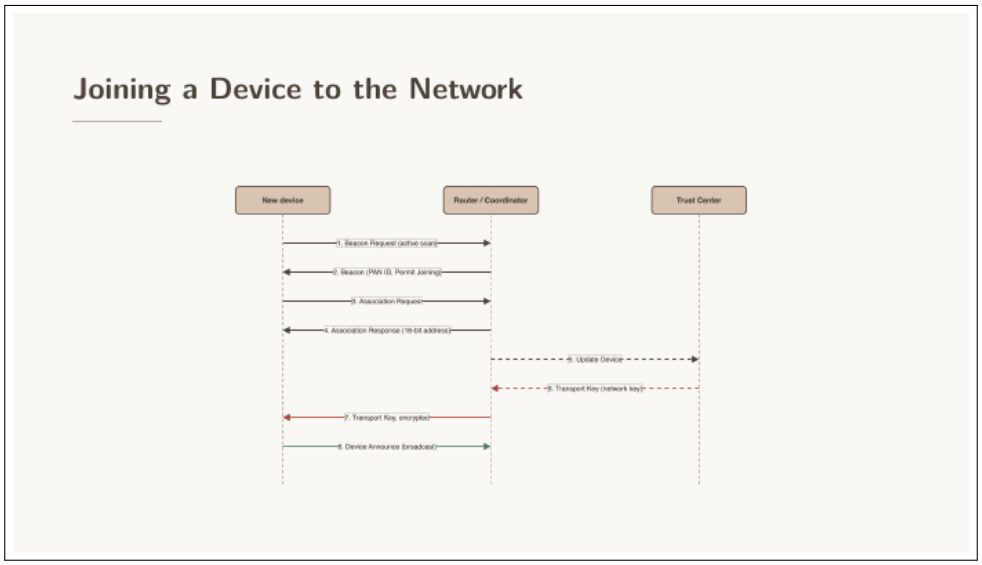


The second part of the mechanisms deals with asymmetry and energy. Links can be asymmetric: each node has its own transmit power and receive sensitivity, so the path from A to B may have a different quality from that from B to A. A high-power node "hears" a weak neighbour that cannot hear it back. If routing assumed symmetry, it would choose paths that work in one direction and fail in the other. That is why the cost of a link considers performance in both directions, and ZigBee accepts as neighbours only bidirectional links.

Energy management is the consequence of what has already been seen about roles. Coordinators and routers are typically mains-powered, because they must be permanently listening. End devices run on battery and sleep. This division is not a convention but a necessity of the architecture: if a router had to sleep, the messages passing through it would be lost.

There is a direct application in installation design. Router positions should be chosen considering the mains connection and coverage. In buildings, lamps and plugs are natural routers. Outdoors, without power, the mesh has to be fed by solar panels or higher-capacity batteries, and the energy budget changes.

Slide 36. Joining a Device to the Network



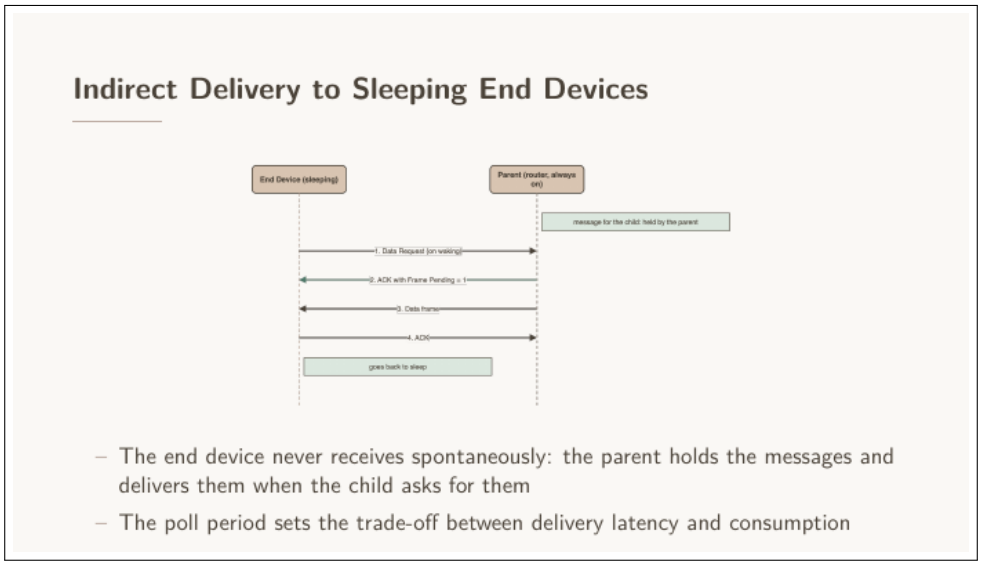
The figure is a sequence diagram of a new device joining a ZigBee network. The three actors are the new device, the router or coordinator it will attach to, and the Trust Center.

The process unfolds in phases. First, the new node performs an active scan: it sends a beacon request (1) on each channel. The routers and the coordinator that allow association answer with a beacon (2) containing the PAN ID and the indication that association is permitted (Permit Joining). The new node chooses the parent, usually the one with the best link, and sends it an association request (3). The parent assigns it the 16-bit short address and answers with an association response (4).

The security phase follows. The parent informs the Trust Center that there is a new device (5). The Trust Center answers with the network key (6), which the parent forwards to the new node (7), encrypted with the link key. Only then is the new node considered authenticated. Finally, the new node announces itself to the network by a broadcast (8), the Device Announce, so that other nodes learn the relation between its 16-bit and 64-bit addresses.

Two practical points. Association is only possible while the parent has permission active, for a limited time; that is why hubs have a pairing mode. And it is in this phase that security is most vulnerable, because the network key is transported: hence the importance of unique link keys, seen later.

Slide 37. Indirect Delivery to Sleeping End Devices



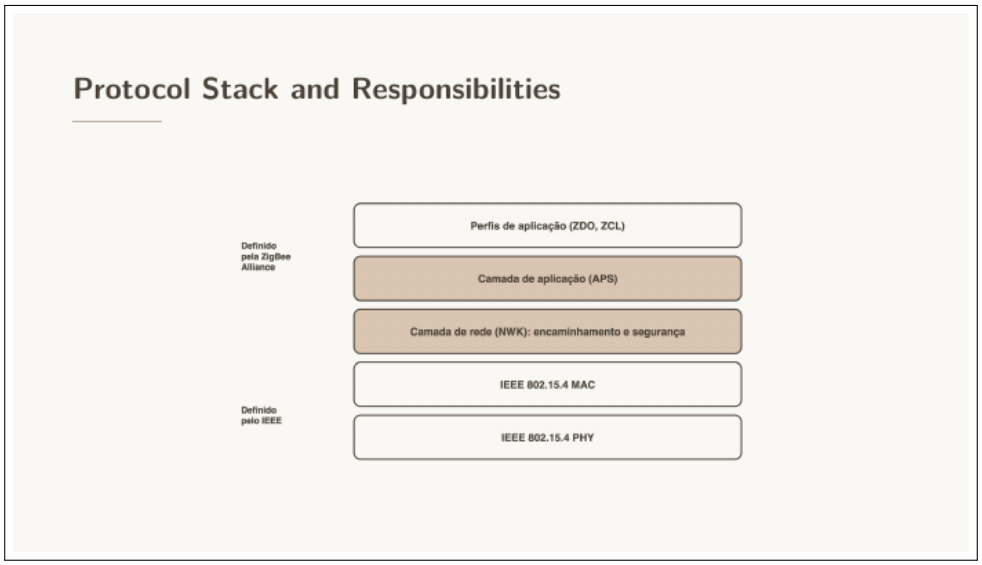
The diagram shows the mechanism that lets end devices sleep without losing messages: indirect delivery. The problem is that a sleeping end device does not receive. If a router or the coordinator wants to send it a message, the end device's radio is off, and the message would be lost.

The solution reverses the initiative. The message is stored at the parent. The end device, on waking, at intervals it decides itself, sends a data request (Data Request) to the parent. The parent answers with an ACK in which the Frame Pending bit is set to 1, indicating that messages are waiting. The end device keeps the radio on, the parent sends the data frame, the end device confirms with an ACK and goes back to sleep. If there are no messages, the ACK carries the bit at 0 and the end device sleeps immediately.

The poll period sets the trade-off between latency and consumption. A 1-second poll allows commands with quick response but uses a lot of energy. A 5-minute poll lasts years on a cell, but a command may take up to 5 minutes to arrive. Sensors that only send data (such as a temperature sensor) can therefore poll rarely or never, and command-driven actuators (a lock) poll more frequently.

The parent stores messages for a limited time and has finite memory. If the end device does not appear, the messages are discarded. In large networks, the number of sleepy children of a router is limited by the parent's buffer space.

Slide 38. Protocol Stack and Responsibilities



The figure presents the protocol stack and the division of responsibilities, and is the slide to return to in order to locate any concept. From bottom to top, the stack has the physical layer (PHY) and the medium access layer (MAC), defined by IEEE 802.15.4. Above are the network layer (NWK) and the application layer, defined by ZigBee. The application layer subdivides into the application support sublayer (APS), the ZigBee device object (ZDO) and the application objects, which implement the profiles.

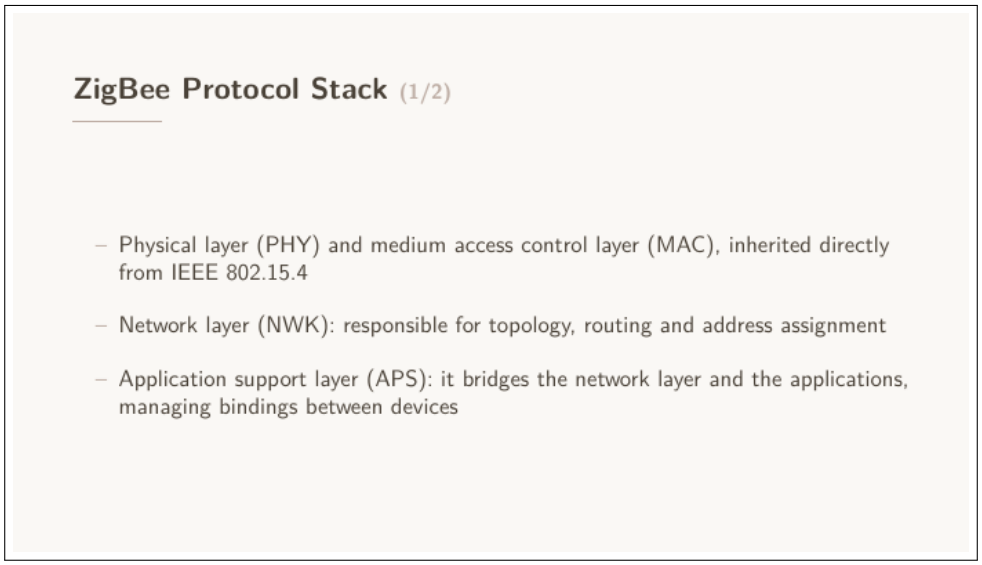
The path of a message illustrates the stack. An application, for example the switch, builds a command according to the profile. The APS adds endpoint and cluster addressing. The NWK adds the network address and routes. The MAC adds the link address and channel access. The PHY converts everything to a radio signal. At the receiver, the path is reversed.

Security cuts across the stack: the NWK layer encrypts and authenticates frames with the network key, and the APS may add protection with link keys.

It is worth comparing with the seven-layer model. ZigBee has no transport or session layer: end-to-end reliability, when present, is provided by the APS with acknowledgements. The stack is deliberately lean, because each layer adds code and bytes.

Check question: state which layers are defined by IEEE 802.15.4 and which by the ZigBee Alliance.

Slide 39. ZigBee Protocol Stack (1/2)



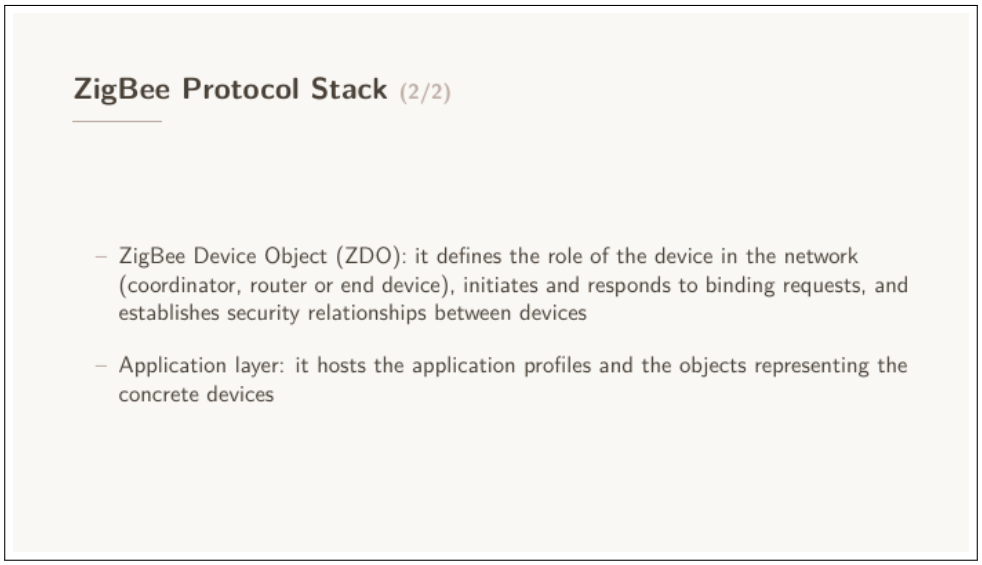
The slide describes the first layers of the stack, focusing on who is responsible for what. The physical layer and the MAC layer are inherited directly from IEEE 802.15.4. The PHY switches the radio on and off, modulates and demodulates, detects energy and assesses the channel. The MAC organises medium access, manages beacons, associations, acknowledgements and, when configured, guaranteed slots.

The network layer (NWK) is the first layer specific to ZigBee and is responsible for three things. Topology: it forms the network, accepts associations, keeps the neighbour list. Routing: it discovers routes, keeps tables and forwards frames. Addressing: it assigns the 16-bit addresses. The NWK also applies security at network level, encrypting and authenticating frames with the network key.

The application support layer (APS) bridges the network layer and the applications. It manages binding tables, which associate an endpoint of one device with one or more endpoints of others, so that a switch can send the command without knowing the lamp's address. It also manages groups, the fragmentation of large messages and end-to-end acknowledgements.

The reason for the division is modularity. A manufacturer writing the application does not need to worry about routing, and the network code can be supplied as a library by the chip manufacturer.

Slide 40. ZigBee Protocol Stack (2/2)



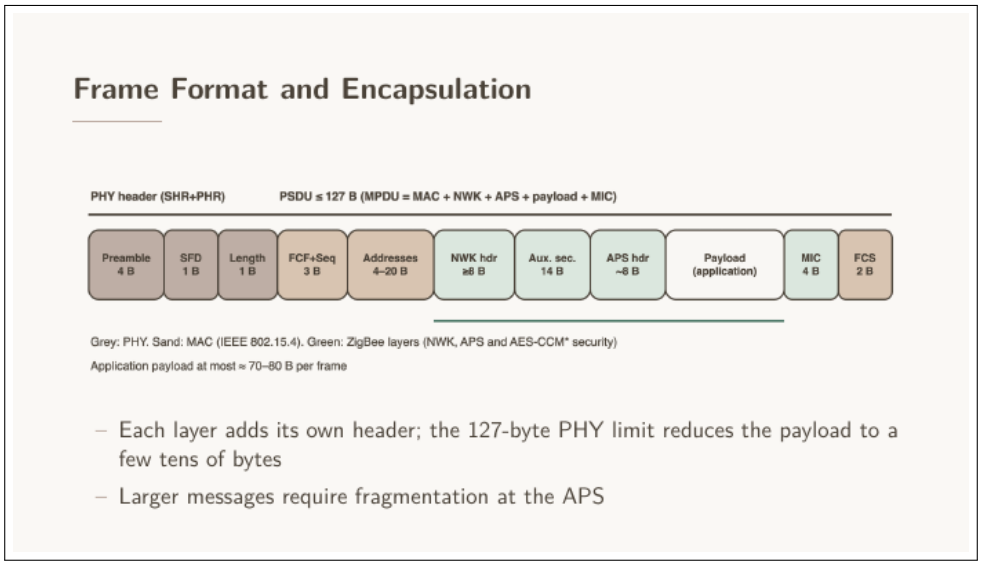
The second part of the stack deals with the ZigBee device object (ZDO) and the application layer. The ZDO is the device manager inside the network. It defines the device's role (coordinator, router or end device), initiates and answers binding requests, discovers the other devices and the services they offer, and establishes the security relationships. It resides on endpoint 0, which is why that endpoint is reserved.

The ZDO discovery services let a newly arrived device introduce itself and the others query it. Typical requests are for the list of active endpoints, the simple descriptor (which profile, which clusters) and matching (who offers this cluster). Thanks to them, a hub that has just paired a lamp knows, without manual configuration, that it supports on/off, dimming and colour.

The application layer hosts the application profiles and the application objects, which represent concrete devices, such as a lamp or a temperature sensor. It is where the manufacturer's logic lives and where the programmer writes code.

The correct reading of the stack is therefore this: IEEE takes care of sending bits to a neighbour; the NWK of carrying them to the destination; the APS and the ZDO of delivering them to the right application and managing the relationship between devices; and the profiles of ensuring both sides understand the data.

Slide 41. Frame Format and Encapsulation



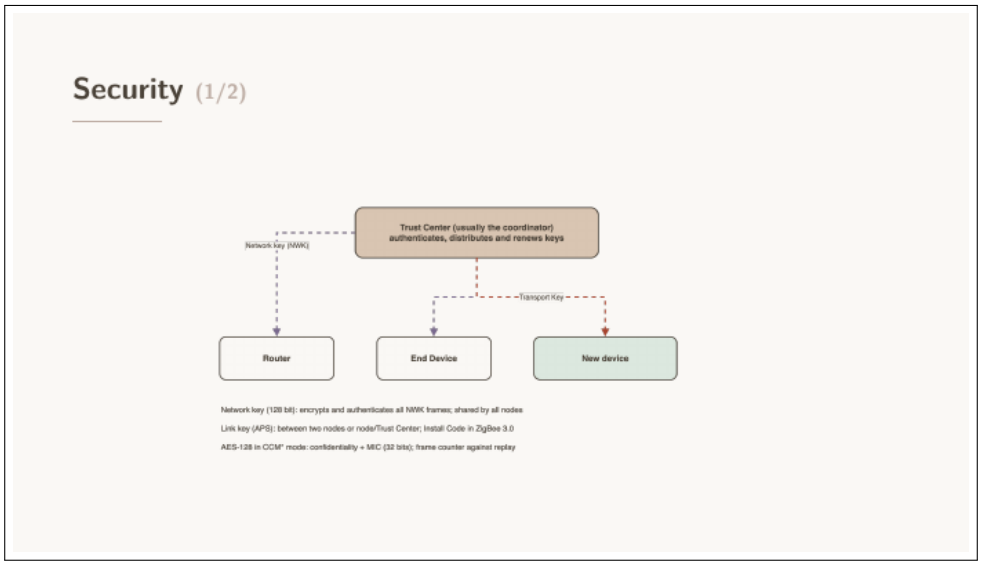
The figure shows how an application message is encapsulated, layer by layer, until it becomes the packet that travels through the air. It is the ZigBee analogue of the TCP/IP encapsulation studied in data networks, but with much more severe space limits.

The sequence, from left to right, is as follows. The PHY adds the preamble (4 bytes), the SFD (1 byte) and the length field (1 byte). The MAC frame follows, with the frame control field, the sequence number and the addresses, 4 to 20 bytes depending on the format. Then comes the NWK frame, with a header of at least 8 bytes containing the source and destination network addresses, the radius and the sequence number. If there is security, the auxiliary security header appears, 14 bytes, with the frame counter. The APS header follows, with the endpoint, cluster and profile. Then the application payload. At the end, the message integrity code (MIC), 4 bytes, and the frame check sequence (FCS), 2 bytes.

The sum of headers, in secured networks, leaves about 70 to 80 bytes for the application within the 127 of the PSDU. A larger message requires fragmentation at the APS level. This is why profiles and clusters use compact values: a temperature attribute takes two bytes, not a text string.

The conclusion for design is that the cost of a byte is not only that of transmitting it: it is radio-on time, energy and collision probability. Headers and payload should be short.

Slide 42. Security (1/2)



The slide presents the ZigBee security architecture. The figure shows the Trust Center, usually the coordinator itself, which authenticates devices, distributes keys and decides who may enter the network. It also shows the network key, shared by all nodes, and the Transport Key with which a new device receives the network key.

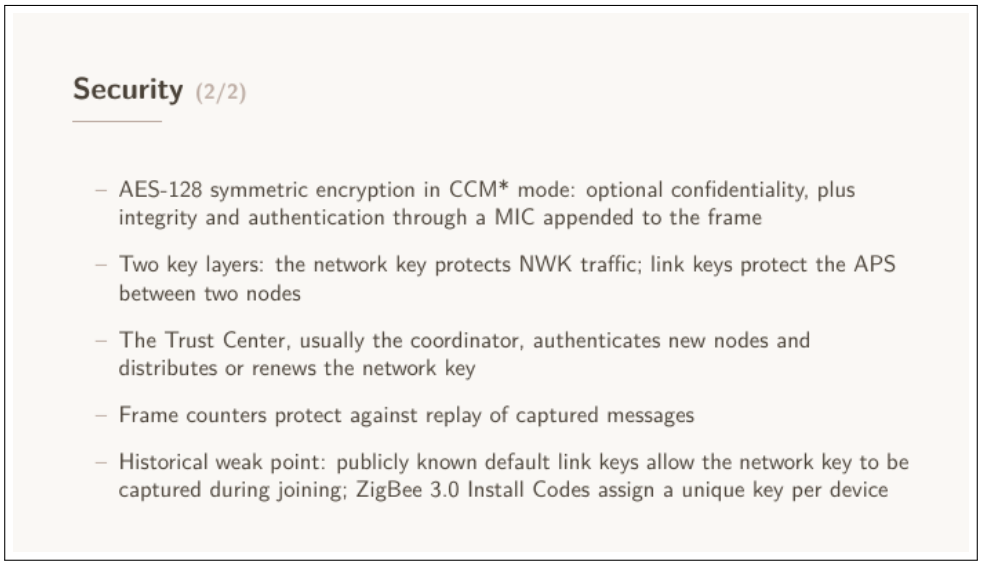
There are two types of key, 128 bits each. The network key is used at the NWK layer: all network frames are encrypted and authenticated with it. Whoever has it can read the traffic and emit valid frames. The link key is used at the APS level, between two nodes or between a node and the Trust Center, and gives additional protection to sensitive messages.

The reason two layers exist is risk management. With a single network key, any compromised node compromises the whole network. Link keys restrict the damage: an attacker who obtains the network key cannot read content protected by link keys.

The cryptographic mechanism is AES-128 in CCM* mode, which combines encryption and authentication. Each protected frame carries an integrity code (MIC) and a frame counter, which prevents a captured frame from being replayed later.

The Trust Center may be centralised, when the coordinator decides everything, or distributed, in which routers may admit new nodes. The first model is more controllable, the second more flexible.

Slide 43. Security (2/2)



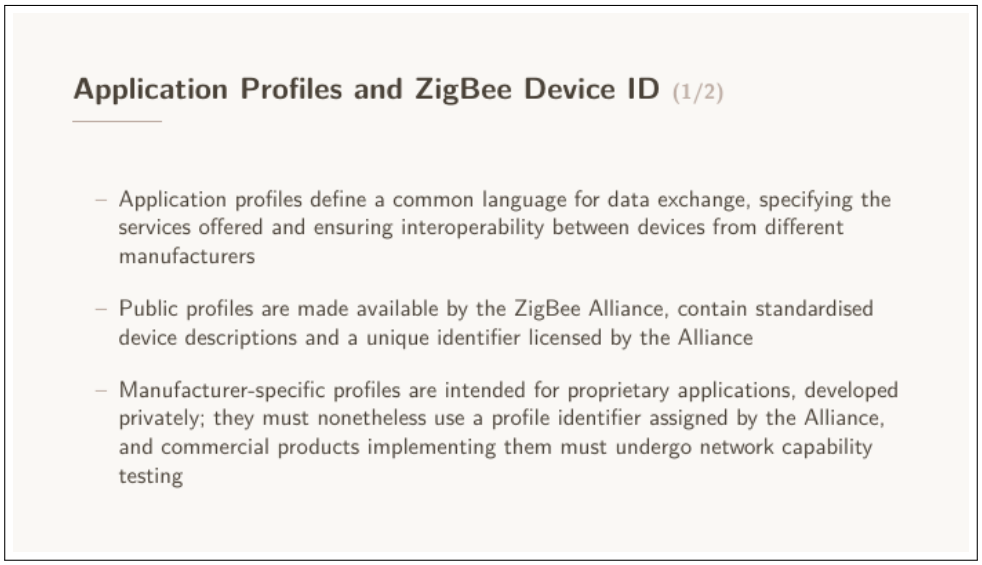
The second part deals with weaknesses and good practice. The fundamental point is that the security of a ZigBee network depends on how the network key reaches the new device. In the early versions, the network key was transported encrypted with a default link key, published in the specification to guarantee interoperability. Whoever knows that key, and it is public, can eavesdrop on the association, capture the network key and from then on read all the traffic. It is ZigBee's historical weak point.

ZigBee 3.0 fixes it with Install Codes: each device comes from the factory with a unique code, printed on the device, from which its own link key is derived. The Trust Center knows the code (entered by the installer, for example by scanning a QR code) and the network key is transported encrypted with a key an attacker cannot guess. The process requires more steps, but it is the recommended practice.

There are other measures. Frame counters protect against replay of captured messages: a frame with a counter lower than the last one seen is rejected. Periodic renewal of the network key limits the exposure time. And the joining window (Permit Joining) should be closed except during pairing.

For Lecture 13, the consequence is practical: Zigbee2MQTT generates a random network key instead of using the default one, and it should not be published or left in code repositories.

Slide 44. Application Profiles and ZigBee Device ID (1/2)



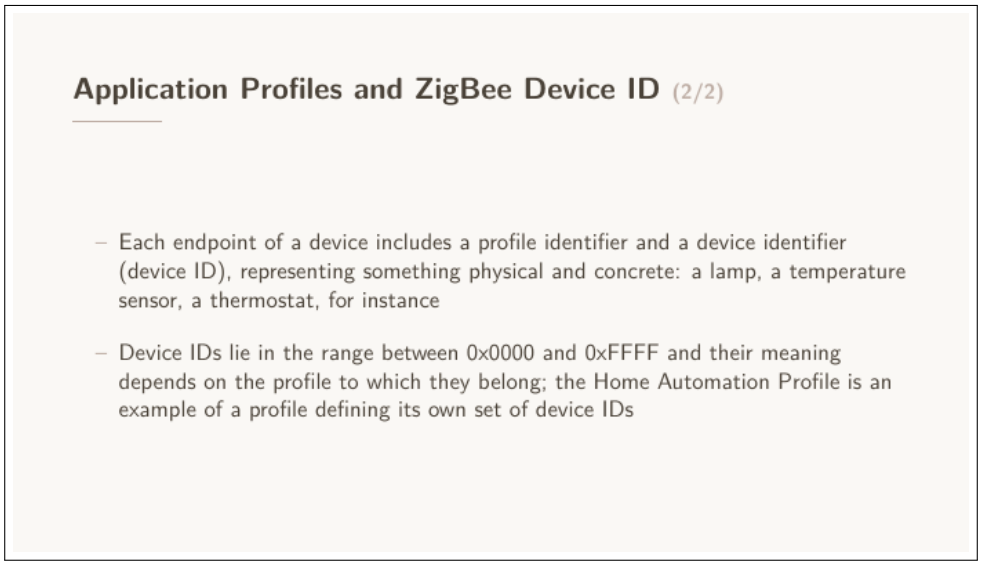
Application profiles are the mechanism that makes ZigBee interoperable across manufacturers. A profile defines a common language for data exchange: what device types exist, what services they offer and what messages are exchanged. Without profiles, a lamp from one manufacturer and a switch from another would share a radio and a network but would not understand each other.

Public profiles are made available by the ZigBee Alliance and contain standardised device descriptions, with a unique profile identifier licensed by the Alliance. Examples: Home Automation, Smart Energy, Light Link, Health Care. Manufacturer-specific profiles are intended for proprietary applications, developed privately, but still have to use a profile identifier assigned by the Alliance. Commercial products implementing them have to be submitted to network capability tests, to ensure they do not disturb other ZigBee networks.

The role of profiles can be seen in practice when buying a certified lamp: the ZigBee seal guarantees that it implements the profile and that, therefore, any certified hub can control it. It is the difference between a technology and an ecosystem.

The cost is rigidity. A manufacturer wanting a feature outside the profile has to use proprietary extensions, and these cease to be interoperable. So, even though profiles exist, the user experience varies between manufacturers.

Slide 45. Application Profiles and ZigBee Device ID (2/2)



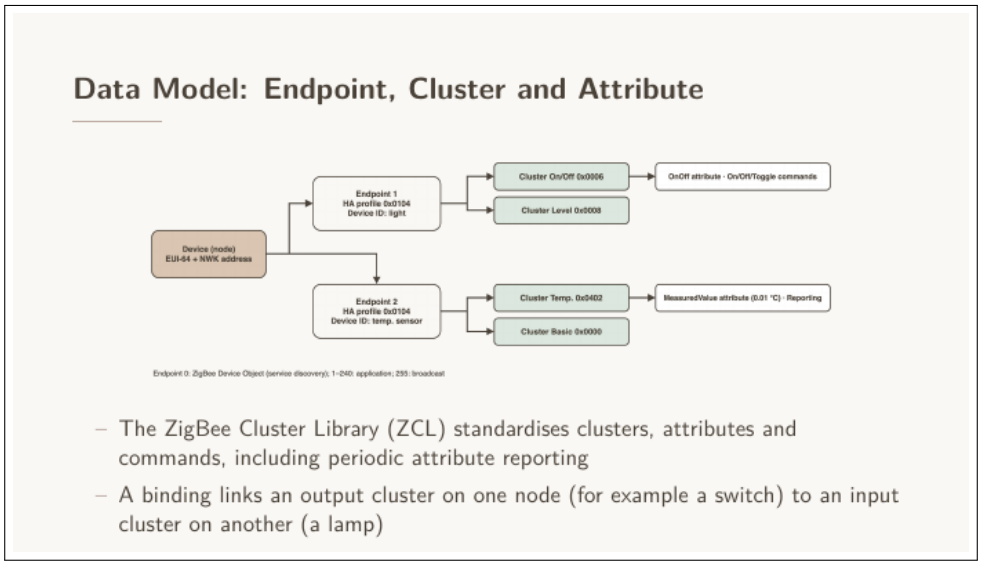
Each endpoint of a device includes a profile identifier and a device identifier (device ID). The device ID represents something physical and concrete, such as a lamp, a temperature sensor or a thermostat, and determines which clusters the endpoint must implement. It is this association between device ID and set of features that allows a hub to treat a lamp from any manufacturer in the same way.

Device IDs lie in the range 0x0000 to 0xFFFF and their meaning depends on the profile. The same numeric value may mean different things in different profiles. The Home Automation profile identifier, for example, is 0x0104, and within it the device IDs distinguish, among others, on/off lights, dimmable lights, plugs and occupancy sensors.

ZDO service discovery uses these identifiers. A hub pairing a device asks it for the list of endpoints and, for each, the profile, the device ID and the clusters. From there it knows what interface to show the user, without anyone having to configure anything.

The relation to retain is the hierarchy: the profile gives the context, the device ID defines the type of object, and the clusters define the available features. The next slide goes deeper into clusters and attributes.

Slide 46. Data Model: Endpoint, Cluster and Attribute



The figure shows the ZigBee data model, the hierarchy in which the application represents the physical world: a device contains endpoints, each endpoint contains clusters, and each cluster contains attributes and commands.

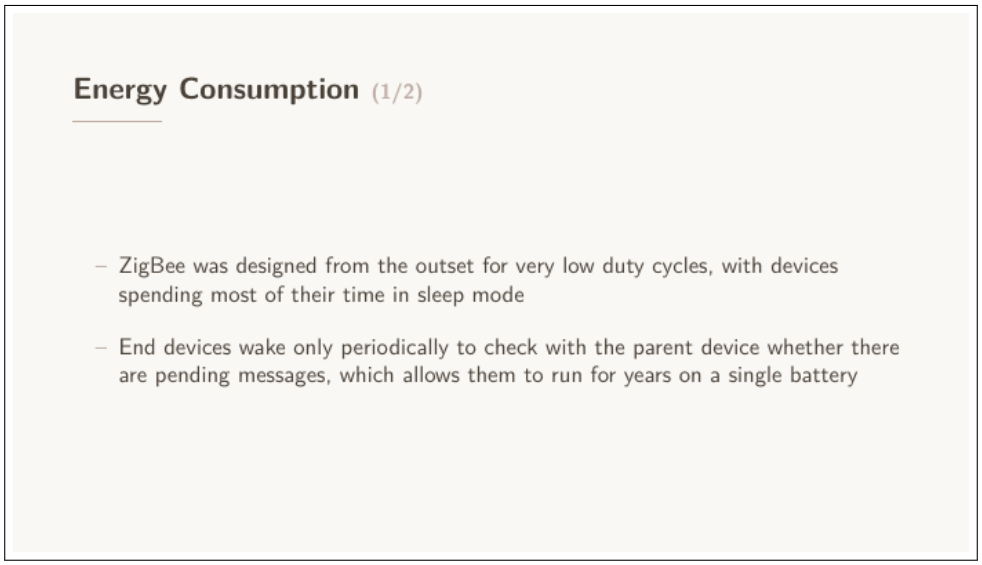
A physical device, identified by the 64-bit address and the network address, may have several endpoints, each with its own profile and device ID. In the figure, the first endpoint is a light and the second a temperature sensor. Each endpoint implements a set of clusters, which are groups of features defined by the ZigBee Cluster Library (ZCL). The On/Off cluster (0x0006) switches on and off; Level Control (0x0008) adjusts intensity; Temperature Measurement (0x0402) supplies measurements; Basic (0x0000) describes the device.

An attribute is a variable of the cluster, for example OnOff, which holds the state, or MeasuredValue, which holds the temperature in hundredths of a degree Celsius. A command is an operation, for example On, Off or Toggle. An attribute can be read, written and, very important for IoT, reported: the device sends the new value spontaneously, periodically or when it changes by more than a threshold, and the receiver does not need to poll.

A binding links an output cluster on one node, for example a switch's, to an input cluster on another, the lamp. Once the binding is created, the switch sends the command without knowing the destination's address.

It is this model that Zigbee2MQTT converts into MQTT messages in Lecture 13.

Slide 47. Energy Consumption (1/2)



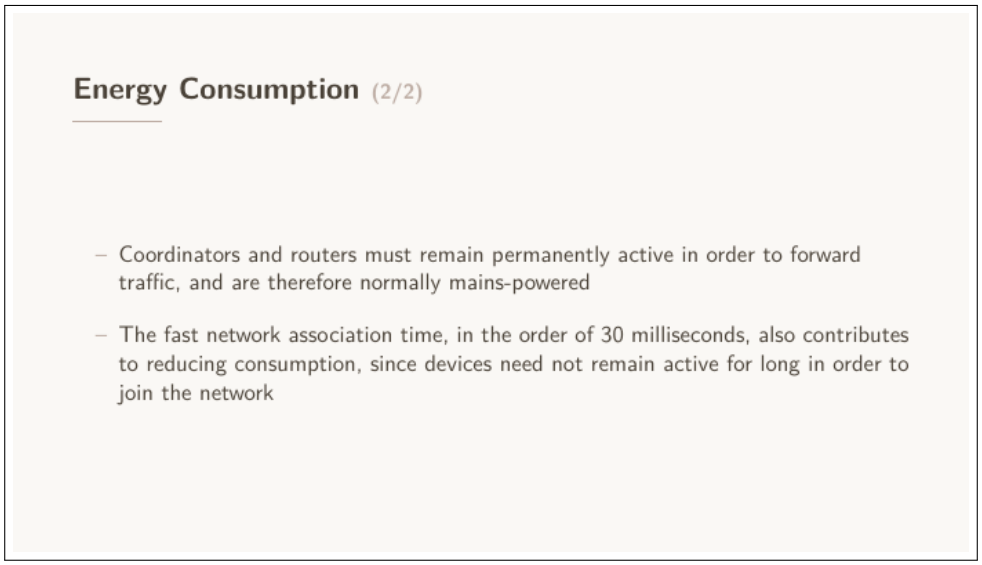
ZigBee was designed from the start for very low duty cycles, with devices spending most of their time in sleep mode. The idea is simple and has deep consequences: the average consumption of a device is dominated by the fraction of time the radio is on, and the current at rest is thousands of times smaller.

End devices only wake periodically to check with the parent whether there are pending messages, the indirect delivery mechanism seen earlier, and to send data when necessary. This logic lets them run for years on a single battery. A CR2032 coin cell has about 225 mAh, and a device with average consumption of a few tens of microamperes lasts a few years.

The asymmetry of radio costs deserves notice. In many modules, transmission and reception draw similar currents, of the order of 25 to 30 mA. Receiving is as expensive as transmitting, so listening to the channel unnecessarily wastes energy. That is why protocol design is so concerned with avoiding permanent listening in battery devices.

The application also influences consumption. A sensor that reports only when the value changes by more than a threshold uses much less than one that reports every second. Application design is thus part of the energy budget.

Slide 48. Energy Consumption (2/2)



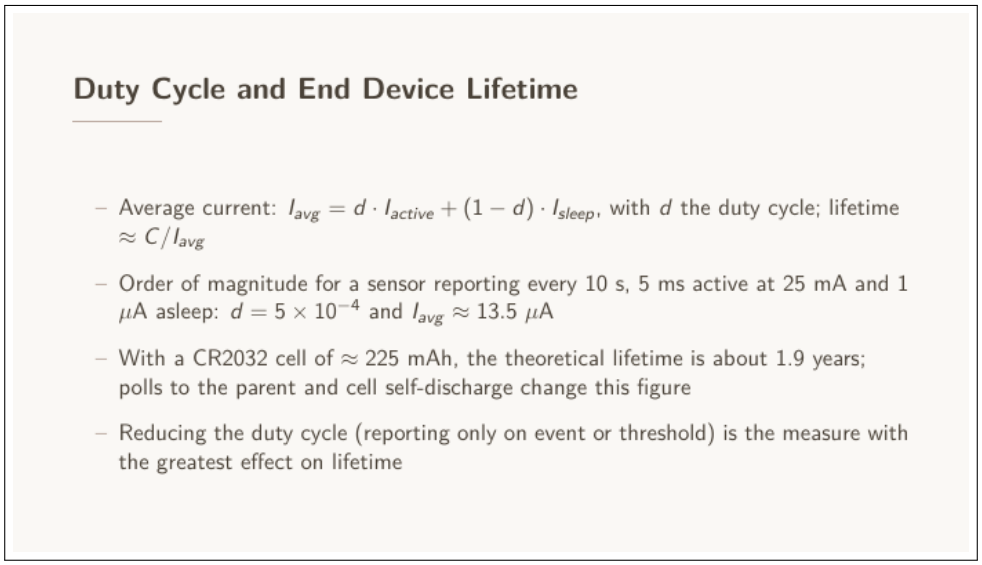
The counterpart is that coordinators and routers must be permanently active to be able to forward traffic, so they are normally mains-powered. A router that slept would stop being reachable, and the routes passing through it would fail. This is why, in a home installation, it is the lamps and plugs, not the sensors, that form the backbone of the mesh.

The slide also mentions the fast association time, of the order of 30 milliseconds, which helps reduce consumption. A device that joins the network in milliseconds can wake, associate, transmit and go back to sleep, spending much less than one needing seconds, as happens in some other technologies. It is a consequence of the stack being light and of association needing few message exchanges.

The energy balance of the network is therefore asymmetric by design: the energy of connectivity is concentrated in the permanently powered nodes, and battery nodes are left with the minimum. Whoever designs an installation must ensure there are enough routers, well distributed and powered, to support the end devices.

When it is not possible to connect to the mains, as in an agricultural field, routers become powered by solar panels or higher-capacity batteries, and the router's energy budget is the limiting factor of network scale.

Slide 49. Duty Cycle and End Device Lifetime



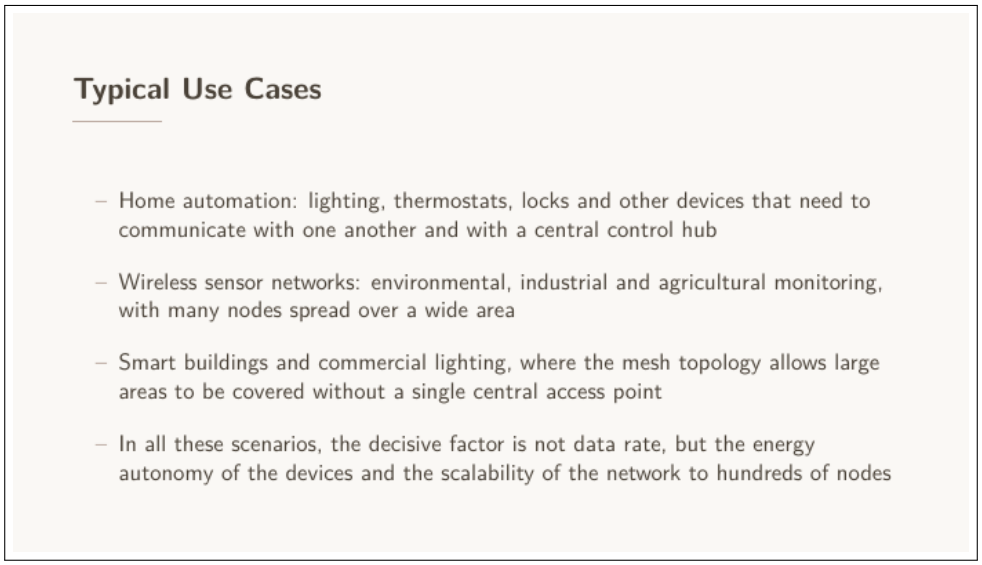
The slide quantifies what the earlier ones state qualitatively. The average current of a device alternating between active and rest is the weighted average of the two currents by the duty cycle d (the fraction of time it is active): $I_{avg} = d \cdot I_{active} + (1 - d) \cdot I_{sleep}$. Lifetime is the battery capacity divided by the average current: C/I_{avg}.

The slide's example: a sensor that reports every 10 seconds, staying active for 5 milliseconds at 25 mA and 1 microampere at rest. The duty cycle is $5\,\text{ms}/10\,\text{s} = 5 \times 10^{-4}$. The average current is $5 \times 10^{-4} \cdot 25\,\text{mA} + (1 - 5 \times 10^{-4}) \cdot 1\,\mu\text{A} \approx 12.5 + 1 = 13.5$ μA. With 225 mAh, the theoretical lifetime is $225\,\text{mAh}/13.5\,\mu\text{A} \approx 16\,700$ hours, about 1.9 years.

A critical reading is essential. This value is an order of magnitude. It ignores polls to the parent, retransmissions, sensor consumption, cell self-discharge and the loss of capacity with temperature and with high peak currents, to which coin cells are sensitive. In practice it may be half or double.

The design lesson is that the dominant term is the duty cycle. Reducing it by a factor of 10, for example reporting every 100 seconds, almost makes the transmission term vanish, and lifetime becomes limited by the rest current and self-discharge. That is why reporting on event or on threshold is the measure with the greatest effect.

Slide 50. Typical Use Cases



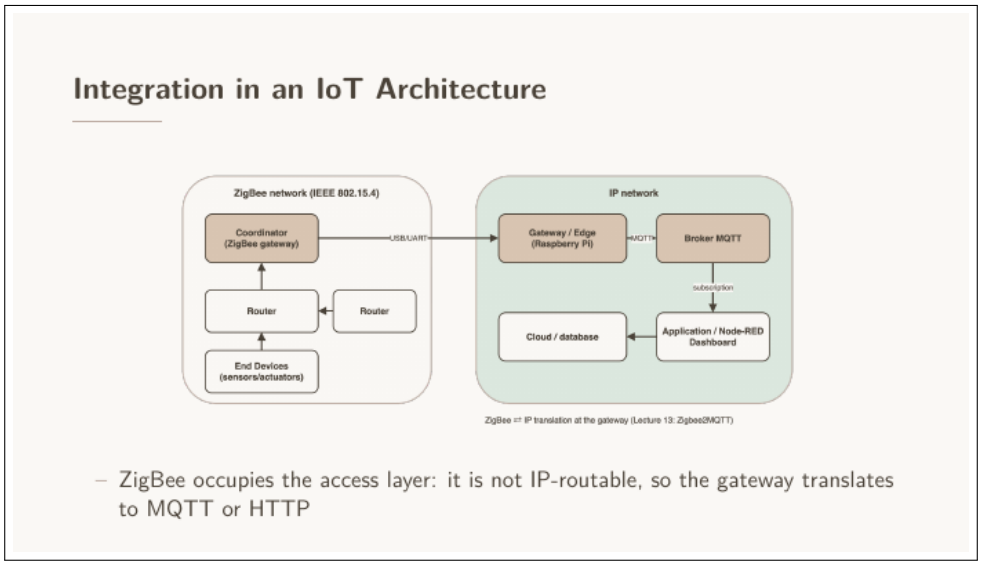
The slide groups the typical use cases into three families and ends with the most important sentence: in all of them, the decisive factor is not the data rate, but the energy autonomy of the devices and the scalability of the network to hundreds of nodes.

Home automation reunites lighting, thermostats, locks and other devices that need to communicate with each other and with a central control hub. Wireless sensor networks cover environmental, industrial and agricultural monitoring, with many nodes spread over a wide area. Smart buildings and commercial lighting use the mesh topology to cover large areas without a single central access point.

To reason about the suitability of a technology for a use case, four questions are worth asking. How many nodes? Scale favours the mesh and compact addressing. What autonomy? If nodes are battery powered, the duty cycle must be very low. What data rate? If bytes per minute, ZigBee is more than enough; if video, it will not do. What coverage? If it exceeds the range of one radio, multi-hop routing is needed.

The next five slides, with a different background colour, make these cases concrete with real examples, indicating the roles of the devices and the topology used in each. When reading them, repeat the four questions and check that the choice of ZigBee is justified.

Slide 51. Integration in an IoT Architecture



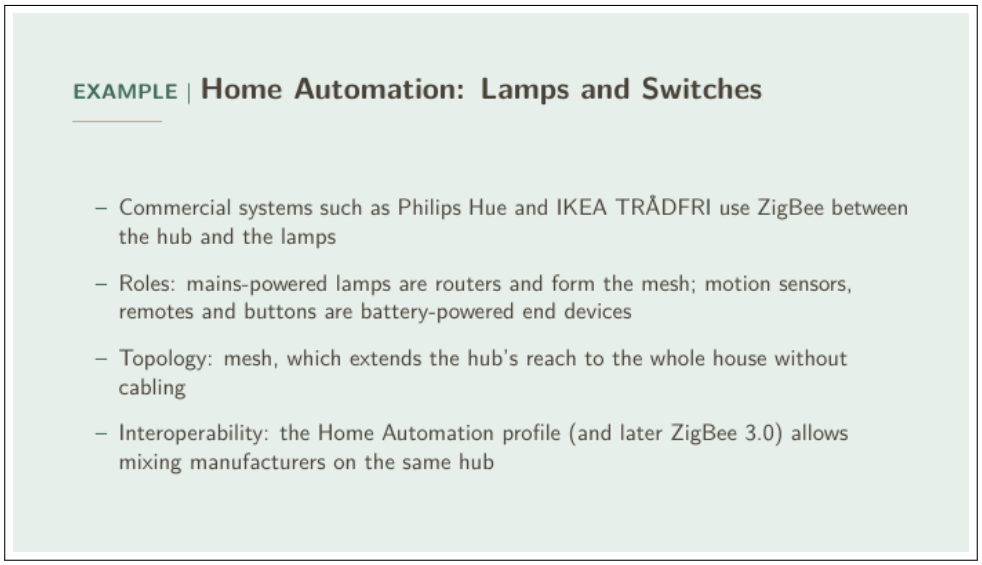
The figure places ZigBee in the architecture of a complete IoT system. On the left is the ZigBee network, with the coordinator, routers and end devices. On the right, the IP network, with the gateway, the MQTT broker, the application and the cloud.

The boundary between the two is the coordinator, connected to the gateway by USB or serial port. The reason is the one seen in earlier lectures: ZigBee devices have no IP address and are not reachable from the Internet. ZigBee is not IP-routable. All communication between the sensor network and the rest of the world passes through a node that speaks both protocols and translates between them.

The gateway, typically a Raspberry Pi, runs software that reads the coordinator's messages, interprets the ZigBee data model (endpoints, clusters, attributes) and publishes them on MQTT topics, in a readable format, usually JSON. In the opposite direction, it subscribes to command topics and converts them into ZigBee commands. The MQTT broker decouples producers and consumers. The application, for example a Node-RED flow or a dashboard, subscribes to the topics and displays, logs or reacts. The cloud allows long-term storage and remote access.

This pattern, with a sensor network upstream, a translating gateway, message-oriented middleware and an application, is what Lecture 13 implements with Zigbee2MQTT. It corresponds to the layered model of Lecture 01: devices, connectivity, platform and application.

Slide 52. exlabel Home Automation: Lamps and Switches



First usage example, marked with the different background colour: consumer home automation. Commercial systems such as Philips Hue and IKEA TRÅDFRI use ZigBee between the hub and the lamps. It is probably the most common contact a person has with the technology without knowing it exists.

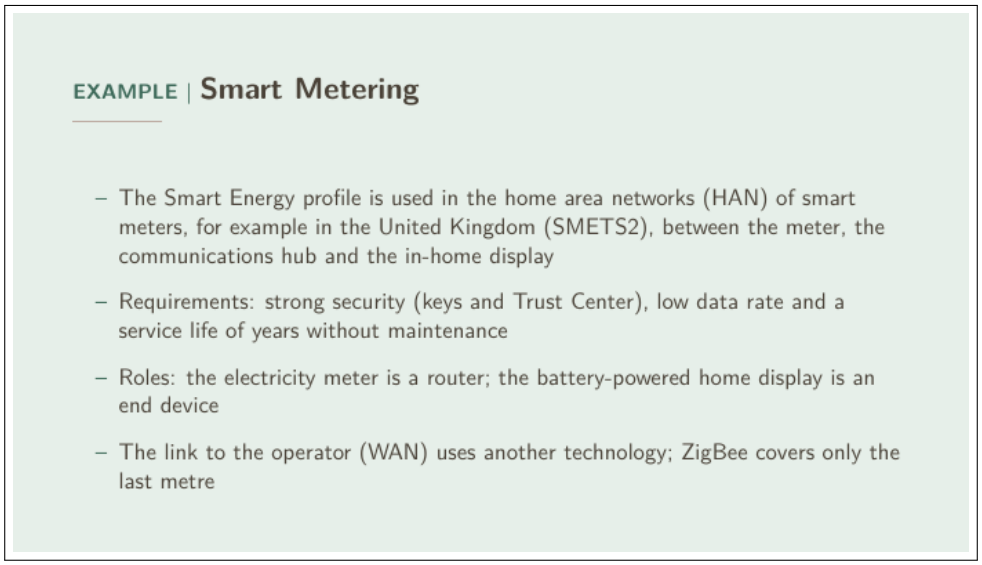
The roles fit the theory. The lamps, mains-powered, are routers and form the mesh: each lamp added to the house makes the network denser and more robust, which is the opposite of what happens with star technologies, where more devices mean more load on a central point. Motion sensors, remotes and buttons, battery powered, are end devices that sleep and only wake to send an event.

The topology is the mesh, which extends the hub's reach to the whole house without cabling. The hub is the coordinator and the bridge to the home network, and from it the phone app or the voice assistant control the lights.

Interoperability results from the profiles, Home Automation and, later, ZigBee 3.0, which allow manufacturers to be mixed on the same hub: a lamp from one manufacturer and a switch from another understand each other because they implement the same clusters (On/Off, Level Control, Color Control).

The reason for choosing ZigBee rather than Wi-Fi is energy and scale: dozens of low-cost devices, without overloading the access point.

Slide 53. exlabel Smart Metering



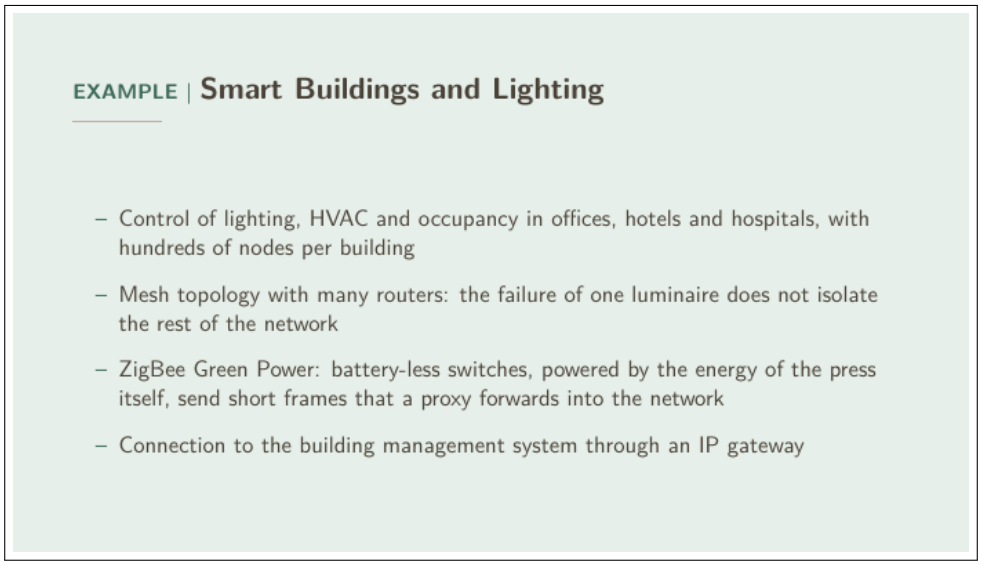
Second example: smart metering. The Smart Energy profile is used in the home area networks (HAN) of smart meters. In the United Kingdom, for example, the second-generation meter infrastructure (SMETS2) uses ZigBee in the link between the meter, the communications hub and the in-home display.

The requirements distinguish this case from the previous one. Security has to be strong, because consumption data are personal and commands could, in theory, switch supply on and off. Hence the use of link keys, a Trust Center and certificates. The data rate is low, because what circulates are readings and price messages. The expected service life is many years, without maintenance, which requires a low-power radio and a stable protocol stack.

The roles: the electricity meter, mains-powered, is a router; the in-home display, battery powered, is an end device. The link to the operator, the WAN, is made by another technology, cellular or dedicated radio: ZigBee covers only the last metre, between equipment in the home.

The example shows a feature that consumer home automation hides: profile standardisation allows a regulator to impose interoperability among manufacturers, so the consumer can change energy supplier without replacing the display.

Slide 54. exlabel Smart Buildings and Lighting



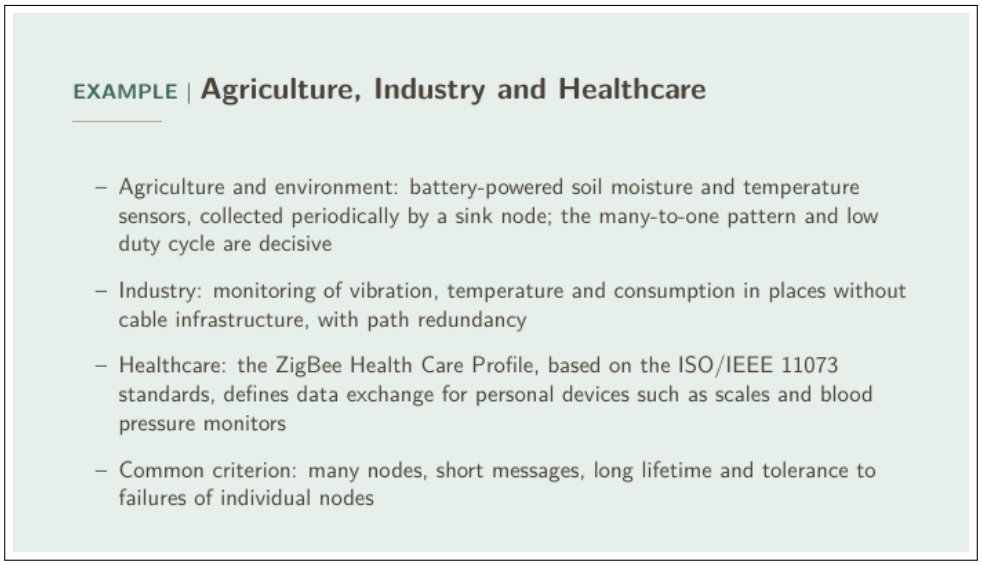
Third example: smart buildings and lighting. In offices, hotels and hospitals, ZigBee is used to control lighting, HVAC and occupancy, with hundreds of nodes per building. The scale is the one the slide on general characteristics refers to.

The topology is the mesh with many routers, and the advantage is the one the lab activity demonstrates: the failure of one luminaire does not isolate the rest of the network, because the neighbours route the traffic along another path. In a building, where maintaining equipment often requires a team, automatic recovery is a requirement, not a luxury.

A particularly interesting extension is ZigBee Green Power, which allows battery-less switches. Instead of a cell, the switch harvests the mechanical energy of the press itself, enough to emit a short frame. That frame is received by a proxy, normally a nearby router, which converts it into normal network traffic. The result is a switch with no cell to replace, which can be stuck on a wall without cabling. It is the combination of low consumption, short frames and a router infrastructure that makes the concept possible.

The link to the building management system is made through an IP gateway, which translates ZigBee messages into the protocols the system uses.

Slide 55. exlabel Agriculture, Industry and Healthcare



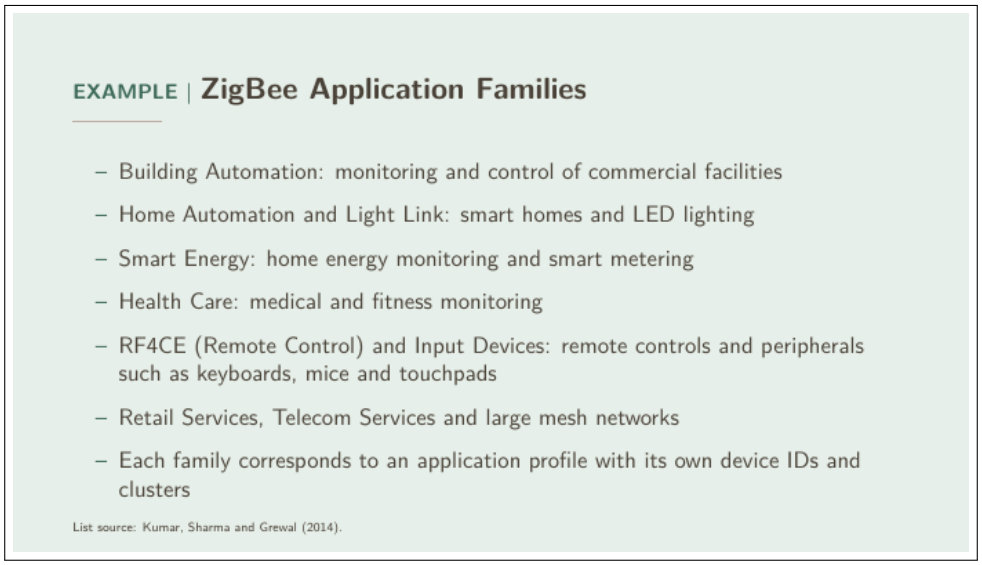
Fourth example, which groups three domains with the same underlying logic. In agriculture and the environment, soil moisture and temperature sensors run on batteries and report periodically to a sink node. The pattern is many-to-one, with all data converging on one point, and the very low duty cycle is decisive: a field has no sockets, and changing the cells of hundreds of sensors is impractical. Coverage requires several hops, and a router mesh, here possibly solar-powered, is the solution.

In industry, ZigBee is used to monitor vibration, temperature and consumption in places without cable infrastructure, with path redundancy. Caution is needed: in industrial environments, interference and noise are high, and requirements for deterministic latency or functional safety fall outside what ZigBee guarantees.

In healthcare, the ZigBee Health Care Profile, based on the ISO/IEEE 11073 standards, defines the data exchange of personal devices, such as scales and blood pressure monitors, with the home gateway. Standardising the data model eases integration with telemonitoring platforms.

The common criterion, which summarises the slide, is many nodes, short messages, long autonomy and tolerance of failures of individual nodes. When a case combines these four traits, ZigBee is a natural candidate; when one is missing, it is worth comparing with the alternatives.

Slide 56. exlabel ZigBee Application Families



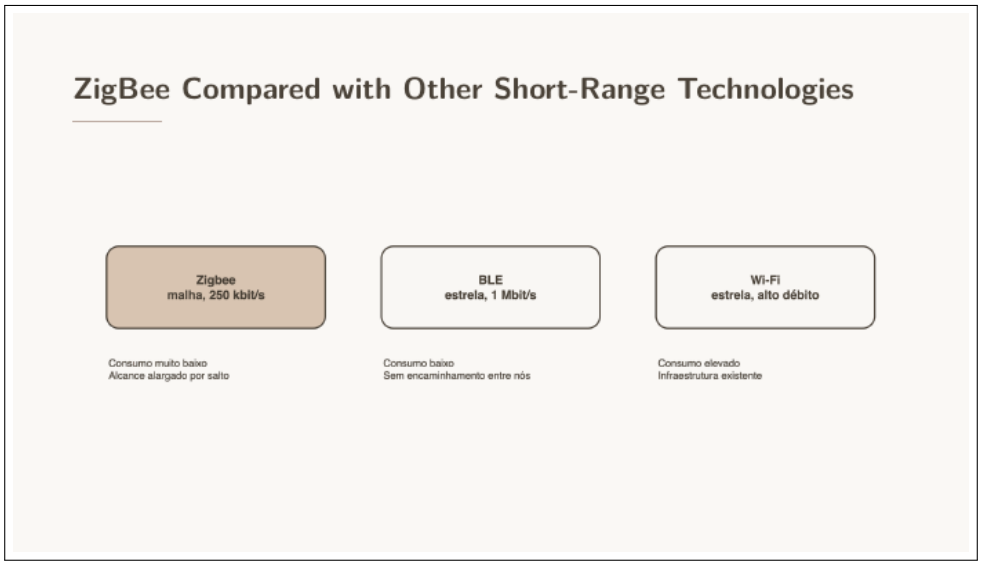
The last slide of examples uses the list of ZigBee application families from the paper by Kumar, Sharma and Grewal (2014). Each family corresponds to an application profile, with its own device IDs and clusters, and is not merely a commercial label.

Building Automation covers the monitoring and control of commercial facilities. Home Automation and Light Link cover smart homes and LED lighting. Smart Energy covers energy monitoring and smart metering. Health Care covers medical and fitness monitoring. RF4CE, radio frequency for consumer electronics, covers remote controls, and Input Devices cover keyboards, mice and touchpads. The list also includes retail services, telecommunications services and large mesh networks.

The useful reading is this. The variety of domains shows that ZigBee is a platform, not an application: the stack is the same, what changes is the profile. It also shows why interoperability is solved at the profile level and not at the radio level: two devices certified on different profiles share the same radio but do not understand each other.

One caveat: the list reflects the state of ZigBee at the date of the paper, 2014. Since then, ZigBee 3.0 has unified several of these profiles in a single specification, and RF4CE has come to be maintained separately.

Slide 57. ZigBee Compared with Other Short-Range Technologies



The slide compares ZigBee with Bluetooth Low Energy and Wi-Fi, the two technologies it is most often confused with or competes against. The comparison runs on three axes: topology, data rate and consumption.

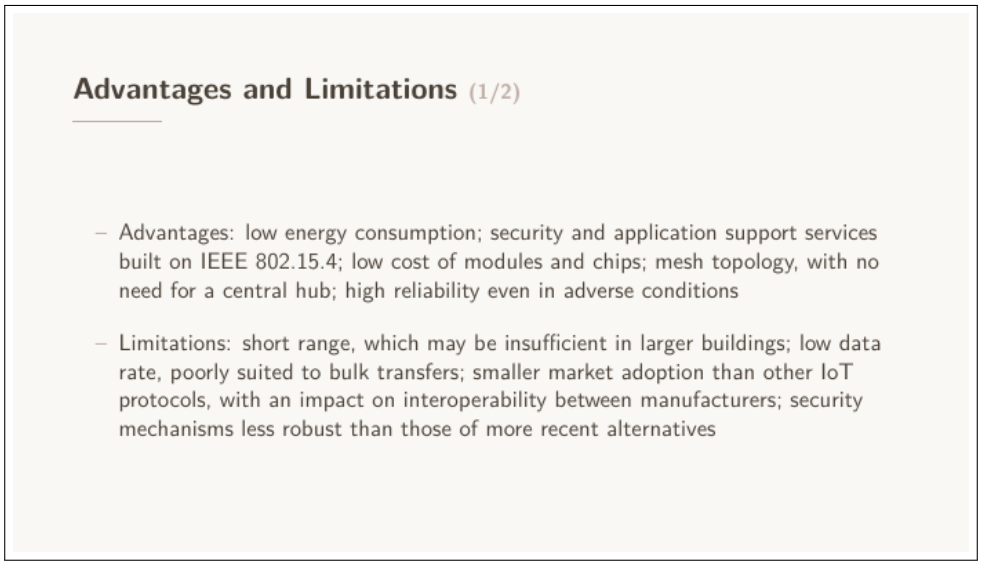
ZigBee is organised as a mesh, with 250 kbps at 2.4 GHz, very low consumption and extended range per hop. Bluetooth Low Energy is organised as a star, with 1 Mbps on the radio, low consumption and, in the classic version studied in Lectures 05 and 06, no routing between nodes. Wi-Fi is organised as a star, around an access point, has high data rate, high consumption and the advantage of reusing existing infrastructure.

Some nuance is needed. Bluetooth today has a mesh variant, Bluetooth Mesh, which makes it comparable to ZigBee in some scenarios, but it is not the one studied. BLE is, on the other hand, natively supported by all phones, which makes it unbeatable when the user interacts directly with the device. Wi-Fi has low-power versions, but its architecture is star and high rate.

The conclusion is that there is no best technology in the abstract. Wi-Fi serves when data rate is needed and infrastructure exists. BLE serves when the device talks to a phone or a nearby gateway. ZigBee serves when there are many battery nodes, spread over an area larger than one radio's range, that need to route among themselves.

Check question: compare the ZigBee topology with the BLE topology studied in Lecture 04, identifying the structural difference.

Slide 58. Advantages and Limitations (1/2)

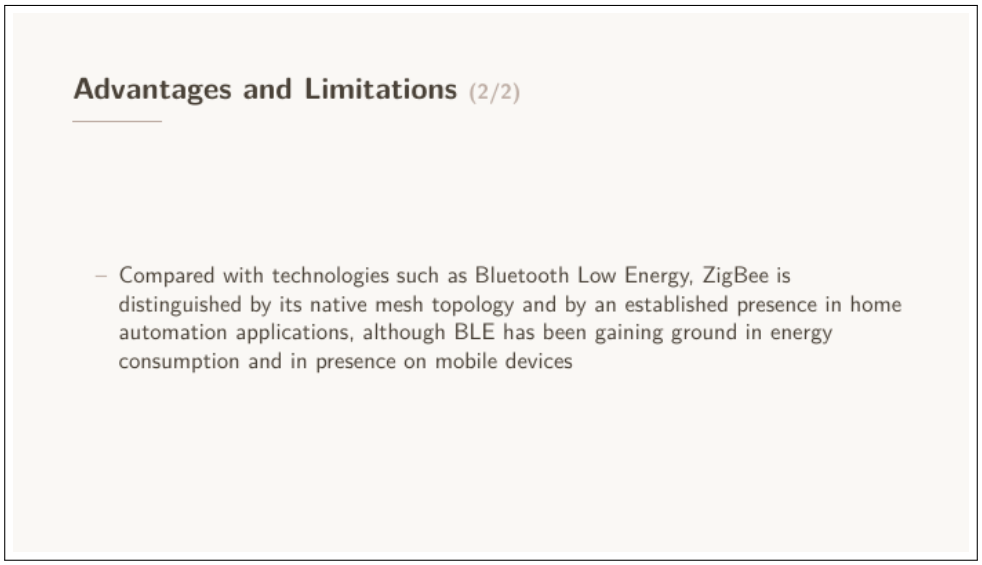


The slide lists the advantages and limitations of ZigBee, and each is worth reading with its cause. The advantages: low energy consumption results from the duty cycle and sleeping; security and application support services come built on IEEE 802.15.4 and AES-128; the low cost of modules and chips follows from the open specification and production volume; the mesh topology dispenses with a central hub, in the sense that there is no single point of failure in routing; and reliability is high, even in adverse conditions, thanks to automatic recovery and frequency agility.

The limitations: the short range may be insufficient in larger buildings, and requires routers; the low data rate makes the technology poorly suited to bulky transfers, such as images or firmware updates (which exist, but are slow); lower market adoption than other IoT protocols has an impact on interoperability among manufacturers, because implementations vary, and some combinations still do not work well; and the security mechanisms, although robust, have historical weaknesses in the joining phase.

The balanced reading is that advantages and limitations are two faces of the same choices. Low data rate and short range are the price of low consumption and low cost. They are not defects to be corrected but trade-offs to be known before choosing the technology.

Slide 59. Advantages and Limitations (2/2)



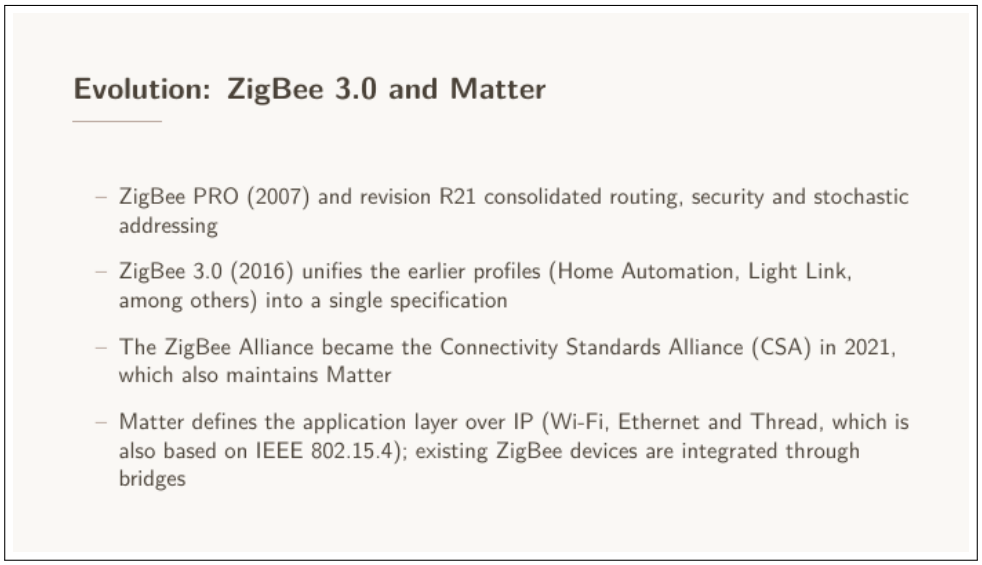
The second part of the slide compares ZigBee with BLE: ZigBee stands out for its native mesh topology and a presence already consolidated in home automation, although BLE has been gaining ground in energy consumption and in presence on mobile devices.

This balance has to be read with concrete data. In consumption, modern BLE achieves very low duty cycles and consumption, and in many applications is comparable to ZigBee or better. In presence, BLE is on every phone, tablet and laptop, which dispenses with a hub for direct interaction; ZigBee always needs a coordinator. In mesh, ZigBee has the advantage of having had it from the start, with a mature product ecosystem, whereas Bluetooth Mesh is more recent.

This explains why the industry has converged on solutions that do not force a choice. Modern home hubs support several radios, typically Wi-Fi, BLE and ZigBee or Thread, and it is the hub, not the user, that manages the diversity. And it explains the emergence of Matter, studied on the next slide.

In design, the question should be asked by requirement, not by fashion. If the device has to talk to the phone without a hub, BLE. If there are hundreds of nodes, spread over a building, battery powered, ZigBee or Thread. If data rate is needed and there is a socket, Wi-Fi.

Slide 60. Evolution: ZigBee 3.0 and Matter

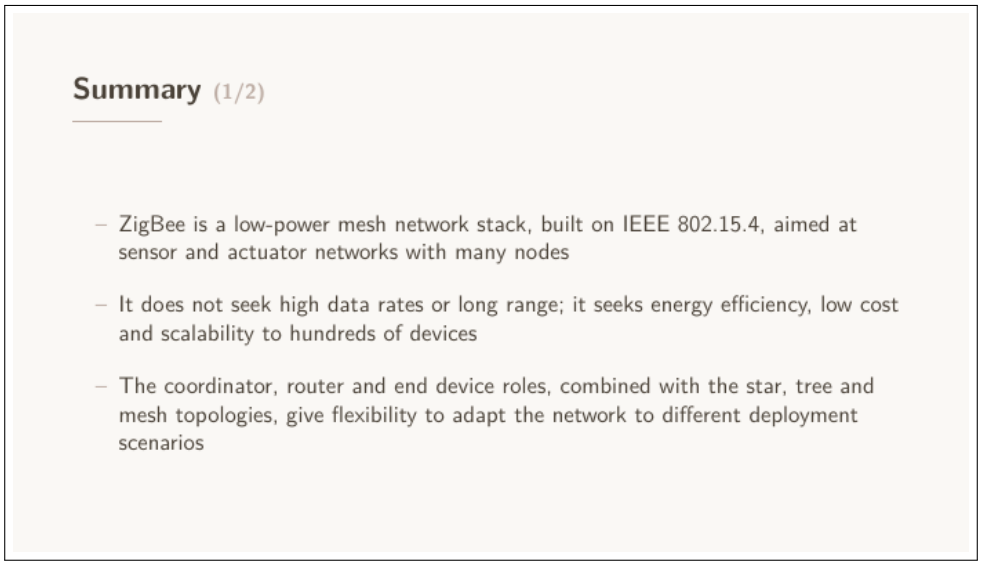


The slide places ZigBee in time. ZigBee PRO, from 2007, and its revision R21 consolidated the routing, security and stochastic addressing studied in this lecture. ZigBee 3.0, from 2016, unified the earlier profiles, such as Home Automation and Light Link, in a single specification, solving a fragmentation problem: before, two ZigBee devices on different profiles could share a radio and not understand each other. In 2021 the ZigBee Alliance became the Connectivity Standards Alliance (CSA), a name that reflects the widening of scope to other technologies.

The CSA also maintains Matter, an application standard over IP for smart homes. Matter does not replace ZigBee at the radio level: it runs over Wi-Fi, Ethernet and Thread. Thread is a low-power mesh network, also based on IEEE 802.15.4, but which uses IPv6 with 6LoWPAN instead of the ZigBee stack, and is studied in Lecture 09. Existing ZigBee devices fit into a Matter ecosystem through bridges, in which a hub exposes the ZigBee devices as if they were Matter.

For the student, the message is twofold. On the one hand, ZigBee is a mature technology, with an enormous installed base, which will continue to exist. On the other, the industry trend is convergence around IP, which is consistent with the direction of the whole course: from the short-range technologies studied here, to the low-power IP networks of Lecture 08 onwards.

Slide 61. Summary (1/2)



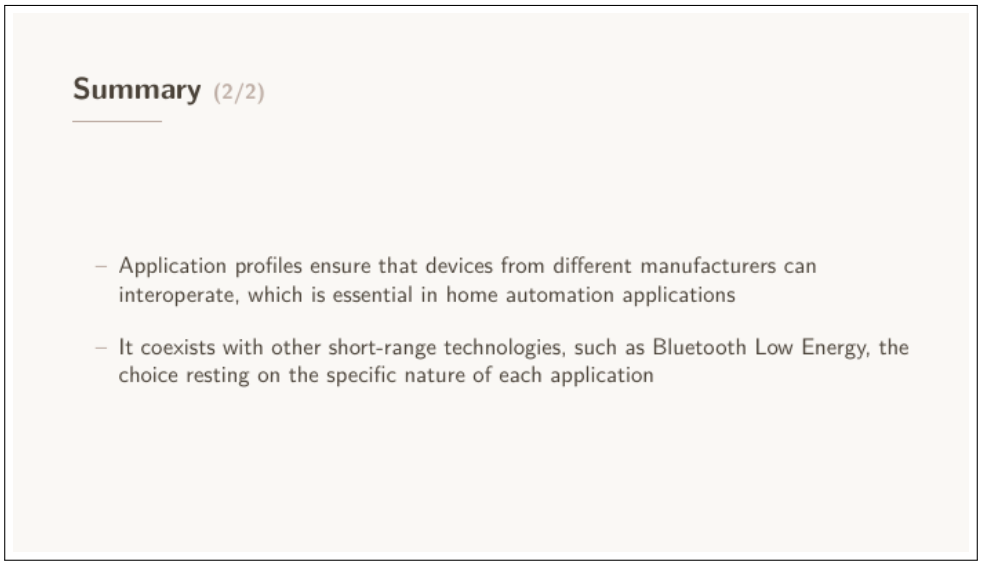
The summary recaps the central ideas. ZigBee is a low-power mesh network stack, built on IEEE 802.15.4, aimed at sensor and actuator networks with many nodes. Each of these expressions refers back to a part of the lecture: network stack (NWK, APS, ZDO), mesh (AODV routing and redundancy), low power (duty cycle, sleeping, indirect delivery), IEEE 802.15.4 (PHY and MAC), and many nodes (16-bit addressing).

The technology does not seek high data rates or ranges. It seeks energy efficiency, low cost and scalability to hundreds of devices. This formulation is the decision criterion: whenever a problem values those three aspects, ZigBee is a candidate; when it values data rate or direct interaction with phones, it is better to look for alternatives.

The roles of coordinator, router and end device, combined with the star, tree and mesh topologies, give flexibility to adapt the network to different deployment scenarios. The asymmetry should be recalled: the coordinator forms the network, routers forward and are always on, end devices sleep. And the topology should be chosen by the trade-off between simplicity and robustness, knowing that mesh robustness protects the routers but not the end devices, which have a single parent.

This summary can be used as a study guide: for each sentence, try to formulate from memory the reasons and mechanisms that support it.

Slide 62. Summary (2/2)



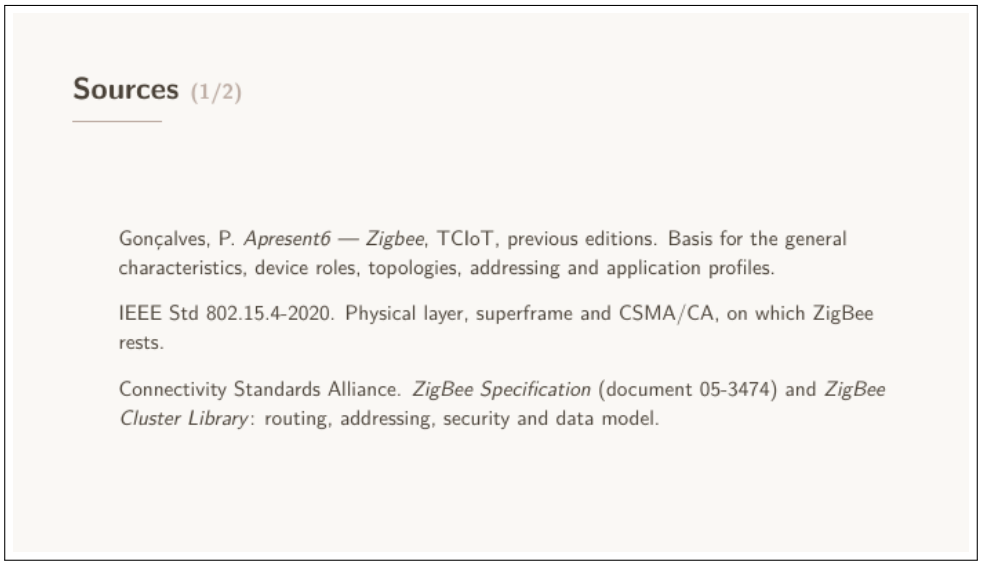
The last points of the summary add interoperability and context. Application profiles ensure that devices from different manufacturers can interoperate, which is essential in home automation. The data model in endpoints, clusters and attributes, fixed by the ZigBee Cluster Library, is what lets a hub treat devices of different origins uniformly, and what Zigbee2MQTT exploits to convert them into MQTT messages.

ZigBee coexists with other short-range technologies, such as Bluetooth Low Energy, the choice depending on the concrete nature of each application. In many real installations the technologies combine: BLE for initial configuration from the phone, ZigBee for the device mesh, Wi-Fi for the hub's link to the Internet.

Before moving on to Lecture 08, check that you can answer the final check questions of these notes. In particular: delimit what belongs to IEEE and what to ZigBee; justify the role of each device type; explain what happens when a router fails in tree and in mesh; justify the gap between 65 000 and 240 devices; explain the existence of two addresses; describe frequency agility; and compare ZigBee with BLE.

Lecture 08 changes level: it leaves the short-range access technologies and enters IP networks for IoT, where IPv4 addressing, NAT and IPv6 explain why the sensor network needs a gateway, and why the trend is to eliminate that need.

Slide 63. Sources (1/2)



The two source slides identify the material on which the lecture rests, and it is good study practice to know it. The basis of the general characteristics, roles, topologies, addressing and profiles is the material from earlier editions of the course. The IEEE 802.15.4-2020 standard defines the physical layer, the superframe and CSMA/CA. The Connectivity Standards Alliance ZigBee specification and the ZigBee Cluster Library define routing, addressing, security and the data model.

For those who wish to go deeper, the recommendation is in order of accessibility. The supporting readings in the lecture folder, the survey by Ramya and others (2015) and the paper by Kaushal and others, give an overview of the topologies, applications and challenges of ZigBee networks. The paper by Kumar, Sharma and Grewal (2014) is a short introduction to the distinction between the standard and ZigBee, useful as a first reading, but to be read critically, because it contains imprecisions, for example in stating that the standard uses only the 2.4 GHz band.

The books by Farahani (2008) and Gislason (2008) are engineering references, with detail on the network layer, routing and product design.

A useful exercise: for each source, identify which slides depend on it, and which claims should be confirmed directly in the specification before being used in a piece of work.

Slide 64. Sources (2/2)



The second sources page includes the more applied references. The usage examples rest on public documentation from manufacturers, such as Philips Hue and IKEA TRÅDFRI, and from the CSA, for the Smart Energy, Green Power and Health Care profiles.

It is important to distinguish the nature of each source when writing a technical report. The standard and the specification are primary sources: they say what the technology is. Scientific papers are secondary sources: they interpret, compare and evaluate, and may contain errors or be out of date. Manufacturer documentation is an application source: it describes what a concrete product does, and has a commercial nature.

Work on the mini-project benefits from this distinction. When justifying a technology choice, it is preferable to cite the standard for operating facts, a paper for performance comparison, and the manufacturer's documentation only for what is specific to the product. The use of artificial intelligence tools to search for information does not remove this care: values and claims must be confirmed at the source, and the source must be declared.

This concludes the expository part of the lecture. The accompanying lab activity, on topologies and resilience, applies the material of slides 23 to 28 and 34 to 37 to a network model, and is the best test of understanding the difference between tree and mesh.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. State which layers are defined by IEEE 802.15.4 and which by the ZigBee Alliance, and give an example of a technology that uses the same radio without being ZigBee.
2. Explain why only the end device can be battery powered and how it receives messages while asleep.
3. Describe what happens to the network when a router fails, in tree and in mesh topology, and why mesh redundancy does not protect end devices.
4. Justify the gap between the 65000 devices of the standard and the 240 units in practice.
5. Explain why each device has two addresses, a 64-bit one and a 16-bit one, and describe the addressing structure (PAN ID, network address, endpoint).
6. State what frequency agility is and what problem it solves; choose ZigBee channels for a network in the presence of Wi-Fi on channels 1, 6 and 11.
7. Describe CSMA/CA and the superframe structure (BO, SO, CAP, CFP), and compute the duty cycle for BO = 6 and SO = 2.
8. Explain route discovery by RREQ and RREP and the role of link cost.
9. Describe the steps by which a new device joins and the role of the Trust Center.
10. Distinguish network key from link key and explain the historical weakness of joining.
11. Describe the device, endpoint, cluster and attribute hierarchy and what a binding is.
12. Estimate the lifetime of a battery sensor from its duty cycle.
13. Compare the ZigBee topology with the BLE topology studied in Lecture 04, identifying the structural difference, and justify the choice of technology for two usage scenarios.

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